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Special Collection:
The Energy Exascale Earth System Model (E3SM): Advancements in a Decade of Earth System Modeling

Key Points:

- Significant updates were made to Earth System Model version 3 atmospheric physics, including gas phase chemistry, aerosols, clouds, and convection
- Improved cloud, convection, and vertical resolution largely improved tropical variability simulation in troposphere and stratosphere
- Improved aerosol representation and aerosol-cloud interactions have led to a much-reduced and realistic aerosol radiative forcing

Correspondence to:
S. Xie,
xie2@llnl.gov

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The Energy Exascale Earth System Model Version 3: 1. Overview of the Atmospheric Component

Shaocheng Xie[1], Christopher R. Terai[1], Hailong Wang[2], Qi Tang[1], Jiwen Fan[3], Susannah Burrows[2], Wuyin Lin[4], Mingxuan Wu[2], Xiaoliang Song[5], Yuying Zhang[1], Mark A. Taylor[6], Jean-Christophe Golaz[1], James J. Benedict[7], Chih-Chieh-Jack Chen[8], Yan Feng[3], Walter M. Hannah[1], Ziming Ke[1,9], Yunpeng Shan[3], Vincent E. Larson[2,10], Xiaohong Liu[11], Michael J. Prather[12], Jadwiga H. Richter[8], Manish Shrivastava[2], Hui Wan[2], Guang J. Zhang[5], Kai Zhang[2], Andrew M. Bradley[6], Philip Cameron-Smith[1], Luis Damiano[6], Bert J. Debusschere[13], Aaron S. Donahue[1], Richard C. Easter[2], Michael S. Eldred[6], Brian M. Griffin[10], Oksana Guba[6], Zhun Guo[10,14], Xianglei Huang[15], Jiwoo Lee[1], Hsiang-He Lee[1], Sijia Lou[2,16], Naser Mahfouz[2], Mitchell Moncrieff[8], Johannes Mülmenstädt[2], Yun Qian[2], Quazi Z. Rasool[2,17,18], Andrew F. Roberts[7], Sean Patrick Santos[2], Khachik Sargsyan[13], Jacob Shpund[2,19], Balwinder Singh[2], Cheng Tao[1], Jinbo Xie[1,20], Yang Yang[2,21], Xubin Zeng[22], Chengzhu Zhang[1], Meng Zhang[1,23], Shixuan Zhang[2], Tao Zhang[4], Xue Zheng[1], Robert L. Jacob[3], L. Ruby Leung[2], Renata B. McCoy[1], and David C. Bader[1]

[1]Lawrence Livermore National Laboratory, Livermore, CA, USA, [2]Pacific Northwest National Laboratory, Richland, WA, USA, [3]Argonne National Laboratory, Lemont, IL, USA, [4]Brookhaven National Laboratory, Upton, NY, USA, [5]Scripps Institution of Oceanography, University of California, San Diego, La Jolla, CA, USA, [6]Sandia National Laboratories, Albuquerque, NM, USA, [7]Los Alamos National Laboratory, Los Alamos, NM, USA, [8]Climate and Global Dynamics Laboratory, U.S. National Science Foundation National Center for Atmospheric Research, Boulder, CO, USA, [9]Now at Division of Atmospheric Science, Desert Research Institute, Reno, NV, USA, [10]Department of Mathematical Sciences, University of Wisconsin-Milwaukee, Milwaukee, WI, USA, [11]Department of Atmospheric Sciences, Texas A&M University, College Station, TX, USA, [12]Department of Earth System Science, University of California, Irvine, CA, USA, [13]Sandia National Laboratories, Livermore, CA, USA, [14]Now at Key Laboratory of Earth System Numerical Modeling and Application, Institute of Atmospheric Physics, Chinese Academy of Sciences, Beijing, China, [15]Department of Climate and Space Sciences and Engineering, University of Michigan, Ann Arbor, MI, USA, [16]Now at Nanjing University, Nanjing, China, [17]Now at Cooperative Institute for Research in Environmental Sciences (CIRES), University of Colorado, Boulder, CO, USA, [18]NOAA Chemical Sciences Laboratory (CSL), Boulder, CO, USA, [19]Now at Department of Atmospheric Science, the Hebrew University of Jerusalem, Jerusalem, Israel, [20]Now at Princeton University, Princeton, NJ, USA, [21]Now at Nanjing University of Information Science & Technology, Nanjing, China, [22]Department of Hydrology and Atmospheric Sciences, University of Arizona, Tucson, AZ, USA, [23]Now at University of Houston, Houston, TX, USA

Abstract This paper describes the atmospheric component of the US Department of Energy's Energy Exascale Earth System Model (E3SM) version 3. Significant updates have been made to the atmospheric physics compared to earlier versions. Specifically, interactive gas chemistry has been implemented, along with improved representations of aerosols and dust emissions. A new stratiform cloud microphysics scheme more physically treats ice processes and aerosol-cloud interactions. The deep convection parameterization has been largely improved with sophisticated microphysics for convective clouds, making model convection sensitive to large-scale dynamics, and incorporating the dynamical and physical effects of organized mesoscale convection. Improvements in aerosol wet removal processes and parameter re-tuning of key aerosol and cloud processes have improved model aerosol radiative forcing. The model's vertical resolution has increased from 72 to 80 layers with the extra eight layers added in the lower stratosphere to better simulate the Quasi-Biennial Oscillation. These improvements have enhanced E3SM's capability to couple aerosol, chemistry, and biogeochemistry and reduced some long-standing biases in simulating tropical variability. Compared to its predecessors, the model shows a much stronger signal for the Madden-Julian Oscillation, Kelvin waves, mixed Rossby-gravity waves, and eastward inertia-gravity waves. Aerosol radiative forcing has been considerably reduced and is now better aligned with community best estimates, leading to significantly improved skill in simulating historical temperature records. Its simulated mean-state climate is largely comparable to E3SMv2, but with some notable degradation in shortwave cloud radiative effect, precipitable water, and surface wind stress, which will be addressed in future updates.

Plain Language Summary This study is part of a series describing the newly released version 3 of the US Department of Energy's Energy Exascale Earth System Model (E3SMv3), focusing on updates to its atmospheric component model (EAMv3). Substantial improvements have been made in representing atmospheric chemistry, aerosols, clouds, convective processes, and their interactions in the model. The model's vertical resolution in the lower stratosphere has increased to better simulate the Quasi-Biennial Oscillation. These updates strengthen E3SM's ability to model aerosol, chemistry, and biogeochemistry, and reduce biases in tropical variability. The model now shows stronger signals for phenomena like the Madden-Julian Oscillation and Kelvin waves. Aerosol radiative forcing is better aligned with community estimates, improving the model's skill in simulating historical temperatures. The model's simulated mean-state climate is largely comparable to its predecessor model EAMv2.

1. Introduction

The US Department of Energy (DOE)'s Energy Exascale Earth System Model (E3SM) project was created to develop a state-of-the-science Earth system model to support energy-relevant science by utilizing DOE high-performance computers (Golaz et al., 2019, 2022; Leung et al., 2020). Since its first version was released in 2018, E3SM has gone through another 5+ years of major model development cycle targeting the implementation of the representation of atmospheric physics, land and energy, land-ice interactions, and coastal waves in its next-generation models, in particular, the third version of E3SM (E3SMv3). This paper is part of a series of overview papers that describe the newly released E3SMv3 with a focus on its atmospheric component and its scientific fidelity based on results from the Atmospheric Model Intercomparison Project (AMIP, Gates, 1992) type of simulations. An overview of its coupled simulations is documented in Part II (Golaz et al., 2025).

The atmospheric physics developments for E3SMv3 atmosphere model (EAMv3) were targeted at the E3SM standard resolution around 100 km in horizontal and focused on the following two major areas: (a) enhancing E3SM capability to simulate the climate response to future scenarios by improving the representation and coupling of aerosols, atmospheric chemistry, and biogeochemistry (BGC) and (b) addressing model shortcomings and deficiencies responsible for major model biases, in particular in precipitation, clouds, aerosol-cloud interactions, and aerosol radiative forcing, exhibited in E3SMv1/v2. The development work as summarized in Table 1 is done through close collaboration across different DOE national laboratories and US universities and research institutes including experts in the identified development areas.

Through nearly a decade of effort, E3SM has become a leading Earth system model used for studying various climate-related topics. Despite the significant advances, EAMv1/v2 (Rasch et al., 2019; Xie et al., 2018) lack an interactive tropospheric gas-phase chemistry and miss or crudely treat several important aerosol species (e.g., stratospheric aerosols, nitrate aerosols, and secondary organic aerosols) that are critical for coupling across aerosol, chemistry, BGC, and other human-Earth systems. The lack of interactive tropospheric chemistry in the previous versions of E3SM has largely limited its applications to study the climate response to various historical emissions and future emission scenarios.

To address this issue, an interactive tropospheric gas chemistry described by Prather et al. (2011) along with an updated linearized stratospheric chemistry scheme (Hsu & Prather, 2010, Linoz v3) were implemented in EAMv3 (Q. Tang et al., 2025). The interactive gas chemistry is coupled with improved aerosol microphysics that include four new features: (a) the addition of a stratospheric coarse mode to the four mode version of the Modal Aerosol Module (MAM4) (Liu et al., 2016; H. Wang et al., 2020), that is, MAM5 with Prognostic Stratospheric Aerosol (MAM5-PSA), for the treatment of stratospheric sulfate aerosol from explosive volcanic eruptions (Hu et al., 2025), (b) a new capability of simulating nitrate aerosol and its radiative forcing through the Model for Simulating Aerosol Interactions and Chemistry (MOSAIC) aerosol chemistry module (Wu et al., 2022), (c) improved representations of Secondary Organic Aerosol (SOA) formation by including sources from gas-phase multigenerational chemistry of SOA precursor gases followed by particle-phase oligomerization and particle-phase SOA chemical sinks due to photolysis (Lou et al., 2020; Shrivastava et al., 2015), and (d) a new scheme to calculate soil erodibility based on the model-predicted soil moisture and fractional areas of the bare soil group, which enables a close coupling of dust mobilization with surface changes (Kok et al., 2014). The improved

Table 1
Summary of the Major Updates in EAMv3 From EAMv2

		Model features	EAMv3	EAMv2
Atmospheric Physical parameterizations	Chemistry	Tropospheric chemistry	UC Irvine interactive chemistry module (Prather et al., 2011)	Specified chemical climatology for troposphere based on other models
		Stratospheric chemistry	Linoz v3 (Hsu & Prather, 2010)	Linoz v2 (Q. Tang et al., 2021)
	Aerosol	Modal Aerosol Module (MAM)	MAM5 with prognostic stratospheric aerosol (Hu et al., 2025)	MAM4 (Liu et al., 2016; H. Wang et al., 2020)
		Nitrate aerosol	MOSAIC (Wu et al., 2022; Zaveri et al., 2008)	None
		SOA	Shrivastava et al. (2015), Lou et al. (2020)	Shrivastava et al. (2015), H. Wang et al. (2020)
		Dust emission	Kok et al. (2014)	Zender et al. (2003)
		Process coupling	Apply surface emissions before turbulent mixing (Wan et al., 2024; Vogl et al., 2024)	Apply surface emissions before dry removal
		Convective transport and aerosol wet removal	Shan et al. (2024), H. Wang et al. (2020)	H. Wang et al. (2020)
		Stratiform cloud microphysics	Predicted Particle Properties (P3) (Morrison & Milbrandt, 2015)	Morrison and Gettelman version 2 (MG2) (Gettelman & Morrison, 2015)
		Minimum in-cloud droplet number concentration	20 cm^{-3}	10 cm^{-3}
	Deep convection	Convective cloud microphysics	Two-moment cloud microphysics scheme (Song & Zhang, 2011; Song et al., 2025)	Simplified microphysics scheme (Zhang & McFarlane, 1995)
		Cloud base mass flux adjustment	Incorporate the large-scale dynamics to determine Cloud base mass flux Song et al. (2023)	Cloud based mass flux depends on CAPE (Zhang & McFarlane, 1995)
		Multiscale Coherent Structure parameterization (MCSP)	Moncrieff (2019)	None
Dynamical Core		Spectral Element (SE) dynamical core on a cubed-sphere grid	Modified hyperviscosity operator for not dissipating large scale impacts of topography (Simmons & Jiabin, 1991) and to add horizontal dissipation to the prognostic variables (Guba et al., 2014)	(Golaz et al., 2022; Taylor & Fournier, 2010; Taylor et al., 1997)
Model Grid and Boundary Condition		Vertical resolution	80 levels	72 levels
		Topography	Six iterations of the Laplace operator	Sixteen iterations (Lauritzen et al., 2015)

representation of aerosols and atmospheric chemistry in EAMv3 has largely enhanced E3SM's capability for coupling aerosols, atmospheric chemistry, BGC, and other human-Earth systems.

Like many other climate models, E3SM exhibits several major model biases in simulating aerosols, clouds, radiation, precipitation, and their interactions although some of the biases are considerably reduced from E3SMv1 to E3SMv2 (Golaz et al., 2022). Examples include but are not limited to the regional biases in the simulated precipitation field (e.g., the dry bias over the central US and Amazon and the excessive precipitation over tropical western Pacific), weak tropical variability (e.g., Madden-Julian Oscillation (MJO), Kelvin waves, and QBO), and incorrect phase and weak amplitude of diurnal cycle of precipitation. Note that most of these biases are long-standing systematic biases that are also present in many other climate models over several model generations (e.g., Bartana et al., 2023; Fiedler et al., 2020; Jiang et al., 2015; Le et al., 2021; J. Lin et al., 2006; H. Lin et al., 2007; S. Tang et al., 2022; Tao et al., 2023). In addition, aerosol effective radiative forcing (ERF_{aer}) in E3SMv1/v2 is too strong compared to community best estimates (Smith et al., 2020), which is thought to be the primary reason for the underestimation of the global mean surface temperature changes in the second half of the historical period (Golaz et al., 2022).

To reduce these model biases, significant model development efforts for EAMv3 have been dedicated at improving the treatment of clouds and convection, as well as aerosol-cloud interactions. Specifically, the Morrison and Gettleman version 2 two-moment bulk microphysics parameterization (MG2, Gettelman & Morrison, 2015) in E3SMv1/v2 was replaced by a new microphysics scheme modified from Predicted Particle Properties

(P3) (Morrison & Milbrandt, 2015) for stratiform clouds to improve the treatment of ice microphysical processes and aerosol-cloud interactions (Shan et al., 2024; J. Wang et al., 2021). The deep convection parameterization, that is, the Zhang-McFarlane deep convection scheme (ZM; G. J. Zhang & McFarlane, 1995), is significantly enhanced through the following improvements: (a) a more sophisticated two-moment bulk cloud microphysics scheme (Song & Zhang, 2011) to replace the simple convective cloud microphysics treatment in ZM for better representing detailed cloud microphysical processes and aerosol-cloud interactions (Song et al., 2025), (b) incorporating dynamical effects of the large-scale vertical motion on convection to determine cloud base mass flux, making convection sensitive to its large-scale environment (Song et al., 2023), and (c) coupling the Multiscale Coherent Structures Parameterization (MCSP, Moncrieff, 2019) with ZM to represent heating associated with mesoscale convective organization (C.-C. Chen et al., 2021). More details about the updates in cloud microphysics and deep convection are described in Terai et al. (2025).

In addition, the treatment of aerosol wet removal processes was more physically consistent in coupling convective cloud properties with wet removal and treating detrained cloud-borne aerosols (Shan et al., 2024), which has led to a considerable reduction of aerosol effective forcing. The dimethyl sulfide (DMS) emission, the number of monolayers for the aging of primary-carbon-mode particles, and the hygroscopicity of particulate organic matter (kPOM) were also re-tuned, and a minimum limiter of 20 cm^{-3} was applied to the in-cloud droplet number concentration, to improve (reduce the magnitude of) ERF_{aer}. As will be shown in this study, the better representation of aerosols, clouds, and convection, as well as their interactions has largely improved the simulation of tropical variability and effective aerosol forcing. The reduced aerosol forcing has led to a significant improvement in simulating the historical surface temperature trend, which is documented in Part II on coupled simulations (Golaz et al., 2025). This has been the most outstanding model bias in E3SMv1 and v2. EAMv3's simulated mean-state climate is largely comparable to EAMv2, with improvements in precipitation and longwave cloud radiative effect. However, there are some degradations in shortwave cloud radiative effect, precipitable water, and surface wind stress, which will be addressed in future updates.

In addition to significant updates in atmospheric physics, the model's vertical resolution in EAMv3 was increased from 72 to 80 layers, with the extra 8 layers added in the lower stratosphere to better resolve stratospheric dynamics. This refined vertical grid, coupled with surrogate-accelerated parameter optimization of the convectively generated gravity wave scheme for parameter retuning, has greatly improved the simulation of the QBO (Damiano et al., 2025; Yu et al., 2025). Furthermore, the coupling of surface emissions, dry removal, and aerosol turbulent mixing was improved to address issues like overly strong dust dry removal, short dust lifetime, and the strong sensitivity of the dust lifecycle to vertical resolution (Vogl et al., 2024; Wan et al., 2024).

Several key diagnostics were also developed or implemented to support EAMv3 model development, including aerosol and aerosol-cloud interaction (ACI) diagnostics for assessing the impact of new parameterizations on aerosol radiative forcing, the E3SM Chemistry Diagnostics Package (ChemDyg) (H.-H. Lee et al., 2025), the tropical variability diagnostics, the Atmospheric Radiation Measurement (ARM) process-oriented diagnostics package (C. Zhang et al., 2020), as well as online conditional sampling and budget analysis capabilities (CondiDiag 1.0; Wan et al., 2022). In addition, the Cloud Feedback Model Intercomparison Project Observation Simulator Package (COSP; Bodas-Salcedo et al., 2011) was updated to version 2 (Swales et al., 2018) with additional MODIS joint histogram diagnostics developed for cloud feedback analysis. A new interface that allows COSP to work with the new cloud microphysics scheme based on P3 was also developed and implemented in EAMv3. COSP improves the comparison of modeled clouds with satellite observations by accounting for observational limitations and instrument features (Y. Zhang et al., 2019, 2024). These new diagnostics have been or are being integrated into the standard E3SM diagnostics package (C. Zhang et al., 2022; K. Zhang et al., 2022).

Finally, EAMv3 includes a few bug fixes to the convective gustiness scheme used in EAMv2. It also contains minor updates to the EAMv2 dynamical core, as well as a rougher topography than EAMv1/v2 that allows for more realistic flow. After incorporating all the new components, minor manual tuning was performed, primarily aimed at improving the top-of-atmosphere (TOA) radiative flux balance, the spatial distribution of precipitation, and both the global mean and spatial distribution of shortwave and longwave cloud radiative effects.

The manuscript is organized as follows: Section 2 provides more details about the major updates of atmospheric physics from EAMv2 to EAMv3. Information about the dynamical core updates, vertical resolution changes, revised surface gustiness formulation for coupling with land and ocean, and adoption of rough topography is also given. Section 3 describes model simulations and observational data used in this study. Section 4 evaluates the scientific fidelity of the model. Section 5 provides an overview of the computational performance of EAMv3. Finally, Section 6 provides a summary of the findings and discusses future directions for E3SM development.

2. Major Updates in EAMv3

EAMv3 uses the same software infrastructure with separate dynamics and physics grids as EAMv2 (see details in Golaz et al., 2022), but it has significantly updated EAMv2 atmospheric physics along with other notable changes in model vertical resolution, the dynamical core, and boundary conditions, which are summarized in Table 1 and described in this section. Only a few parameterizations were not changed during the EAMv3 development. These include (a) the Rapid Radiative Transfer Model for general circulation models (RRTMG) (Iacono et al., 2008; Mlawer et al., 1997) for radiative transfer; (b) the gravity wave parameterization described in Richter et al. (2010); and (c) the Cloud Layers Unified By Binormals (CLUBB) parameterization (Golaz et al., 2002; Larson, 2017; Larson & Golaz, 2005) for turbulence, shallow convection, and macrophysics of stratiform clouds.

2.1. Atmospheric Physical Processes

2.1.1. Atmospheric Chemistry

Partly due to the computational concern, EAMv1 and v2 do not have an interactive chemistry in the troposphere. To address this gap, EAMv3 implemented the UC Irvine interactive chemistry module (chemUCI) described in Prather et al. (2011) in the troposphere. The chemUCI mechanism includes 28 advected tracers for the O_3-CH_4-HO_x-NO_x-NMVOCs chemistry (CH4): methane; ($HO_x \equiv H + OH +$ peroxy radicals; $NO_x \equiv NO + NO_2$; NMVOCs: non-methane volatile organic compounds). Enabling E3SM's applications to study the climate response to various short-lived climate forcers.

In the stratosphere, the linearized stratospheric chemistry scheme (Linoz v2) (Q. Tang et al., 2021) in EAMv2 was updated to Linoz version 3 (Linoz v3) (Hsu & Prather, 2010). The Linoz v3 model calculates four prognostic tracers (O_3, N_2O, NO_y—reactive nitrogen oxides, and CH_4) and diagnoses water vapor. These species are all linked to the chemUCI variables, which allows for seamless chemistry-transport modeling throughout stratosphere and troposphere. Linoz v3 can be driven with emissions of CH_4 and N_2O, along with tropopause boundary conditions for H_2O for full stratosphere-troposphere coupling. The tropopause definition follows the tracer-based E90 tropopause (Prather et al., 2011). More details of atmospheric chemistry are documented in the chemistry overview (Q. Tang et al., 2025).

The main capabilities of the new chemistry module include: (a) fully interactive chemistry in both troposphere and stratosphere, which includes all SLCFs; CH_4 chemistry and lifetime; (b) oxidants for production and photolytic loss of particle-phase SOA_s; oxidation of NO_x to nitrate and SO_2 to sulfate; (c) a diagnosed stratosphere-troposphere that responds to climate change. These capabilities are critical for coupling chemUCI and BGC for ozone and methane.

2.1.2. Aerosol

EAMv3 includes several important updates to EAMv2 aerosol-related processes or features. They are described below.

2.1.2.1. Prognostic Stratospheric Sulfate Aerosol

Instead of prescribing stratospheric aerosol optical properties, like in EAMv2 and many other climate models, EAMv3 now includes a prognostic treatment of stratospheric sulfate aerosol. The four-mode version of the Modal Aerosol Module (MAM4) used in EAMv2 for predicting the number and mass concentrations of major aerosol species (Liu et al., 2016; H. Wang et al., 2020) has been updated to a 5-mode version of the Modal Aerosol Module with Prognostic Stratospheric Aerosol (MAM5-PSA) with a fifth mode specifically designed to represent stratospheric coarse sulfate particles primarily originating from explosive volcanic eruptions (Hu et al., 2025). The fifth mode has a narrower size distribution than the existing coarse mode to better represent the gravitational settling velocity of volcanic sulfate particles without degrading the representation of dust and sea salt particles in the coarse mode (Visioni et al., 2022). The added mode allows EAMv3 to better represent stratospheric sulfate aerosol radiative impact and simulate natural climate forcing associated with volcanic eruptions as well as stratospheric sulfur injection as a geoengineering approach to counteracting global warming.

2.1.2.2. Nitrate Aerosol

As nitrate aerosols play an increasingly important role in affecting regional air quality as well as Earth's climate, EAMv3 incorporates the MOSAIC aerosol chemistry module (Zaveri et al., 2008) to explicitly simulate the spatiotemporal distribution of nitrate and ammonium aerosols and their radiative effects (Wu et al., 2022). MOSAIC is a comprehensive aerosol chemistry module for simulating the dynamic partitioning between gases (e.g., H_2SO_4, semivolatile inorganics HNO_3, NH_3, HCl) and particles of different sizes and compositions, which can replace the default MAM4 treatment of gas-aerosol exchange in EAMv3. The aqueous (i.e., cloud water) chemistry, which already includes uptake and aqueous-phase oxidation of SO_2, is also modified to include uptake of HNO_3, NH_3, and HCl gases. New aerosol species (NH_4, NO_3, Ca, CO_3, Na, Cl) are introduced to the Aitken, accumulation and coarse modes of MAM4(5). MOSAIC is also coupled with chemUCI gas chemistry that predicts HNO_3 through O_3-NO_x-HO_x chemistry or N_2O_5 hydrolysis. The simulation of nitrate aerosols by MOSAIC strongly depends on the simulation of HNO_3 by gas chemistry scheme (Wu et al., 2022, 2025). Note that, due to its computational cost, which makes EAMv3 around 30% more expensive, MOSAIC is currently not set as a default feature for the standard EAMv3 configuration, but it can be turned on for research purposes.

2.1.2.3. SOA

EAMv2 simply uses a single condensable organic gas precursor (called SOAG) and a single semivolatile SOA aerosol species in the different size modes of MAM4. EAMv3 improves the treatments of SOAG and SOA using a modified volatility basis set (VBS) approach to simulate SOA formation explicitly by including multigenerational aging via fragmentation and functionalization reactions, and particle-phase oligomerization, as well as chemical loss of SOA by photolysis (Lou et al., 2020; Shrivastava et al., 2015). The SOAG emissions sources in EAMv3 represent organic gas emissions, and include natural biogenic, biomass burning, and anthropogenic organic gases that are oxidized to form lower volatility species and subsequently undergo gas-particle partitioning to form SOA. The gas-particle partitioning of each semi-volatile VBS species is calculated dynamically, and local non-advected particle-phase tracers for given VBS species in each relevant size mode are invoked for these calculations. Therefore, this scheme represents the chemical complexity of gas-phase multigenerational chemistry forming SOA while maintaining computational efficiency.

2.1.2.4. Dust Emission

One of the main deficiencies in the EAMv2 dust scheme is that the dust emissions depend on the time-invariant soil erodibility, which is calibrated with the present-day climatology (Zender et al., 2003). To address this issue, EAMv3 implements a new dust scheme (Kok et al., 2014) that calculates the time-varying soil erodibility factor for dust mobilization based on the E3SM Land Model version 3 (ELMv3) predictions of surface conditions, such as soil moisture and bare-ground fraction, thus enabling a close coupling of land and atmosphere through dust-climate feedback. The new scheme also includes dust sources in high latitudes (e.g., poleward 60°N), which account for up to about 3% of global dust emissions in the present-day conditions. The high-latitude dust could be an additional ice nucleation source for mixed-phase clouds in polar regions (Shi et al., 2022) and impact the high-latitude ocean BGC through the enhanced supply of iron nutrients from dust (Hamilton et al., 2020).

2.1.2.5. Aerosol Wet Removal

The excessive aerosol forcings in E3SMv2 (Golaz et al., 2022) are thought to be partly related to insufficient aerosol wet removal (Shan et al., 2021, 2023), which is the dominant sink for accumulation mode aerosols (Liu et al., 2012). Several improvements are made to the representation of the aerosol wet removal processes in EAMv3 (Shan et al., 2024): (a) Improving the coupling of cloud properties from the newly implemented convective cloud microphysics (ZMmicro, discussed in Section 2.1.3) with the deep convective wet removal parameterization used in EAMv1 and v2 as described in H. Wang et al. (2020). Specifically, the prescribed deep convective precipitation efficiency for calculating aerosol wet removal rate is now interactively diagnosed from ZMmicro. Similarly, the prescribed in-cloud supersaturation for secondary aerosol activation is calculated based on the convective vertical velocity from ZMmicro. (b) Improving the evolution of convective cloud-borne aerosols by detraining them into stratiform clouds to allow for further removal by stratiform precipitation. Additionally, a scaling factor of 0.6 that was used to adjust the stratiform cloud removal rate is now removed because the new cloud scheme based on P3 predicts less cloud liquid, which leads to a weaker stratiform aerosol

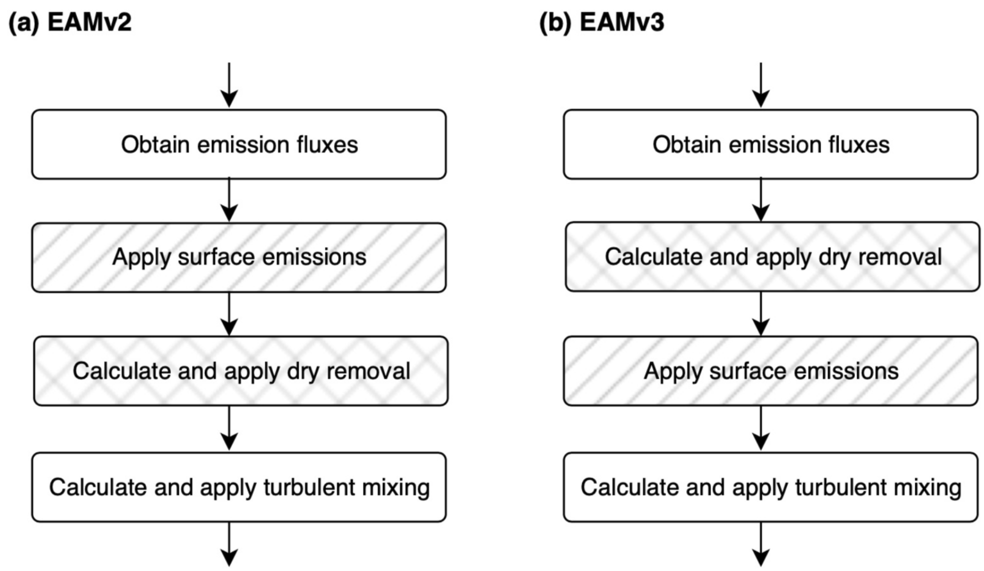

Figure 1. Simple schematics illustrating the key difference between EAMv2 and EAMv3 in terms of the numerical coupling of aerosol-related processes. A more detailed schematic depicting the difference in relation to other physical processes can be found in Figure 1 of Wan et al. (2024).

wet removal. (c) Removing the inconsistency in the coupling sequence for the calculation of convective/stratiform clouds and their wet removals in EAM. More details are documented in Shan et al. (2024).

2.1.2.6. Numerical Coupling of Aerosol-Related Processes

A revised numerical scheme is implemented for the coupling of aerosol-related processes. In EAMv1 and v2, the sequence of calculations was such that the surface emissions of aerosols were applied, then dry removal was calculated and applied, and turbulent mixing was calculated and applied later in the time loop.

As illustrated in Figure 1, in the revised scheme, surface emissions of aerosols are applied before turbulent mixing (instead of before dry removal). As a result, the overly strong surface dry deposition fluxes of dust in EAMv1 (Y. Feng et al., 2022; Vogl et al., 2024; Wan et al., 2024) decrease substantially in source regions (for numerical reasons) and increase substantially outside source regions (because of higher dust loading). Surface wet deposition fluxes of dust increase significantly (global annual mean + 28% in a 5-year present-day AMIP simulation). It also largely reduces the strong sensitivity of the dust lifecycle to vertical resolution in earlier versions of EAM (Wan et al., 2024).

2.1.3. Cloud Microphysics

2.1.3.1. P3 for Stratiform Clouds

EAMv3 replaced MG2 with P3 for stratiform clouds. The use of P3 is to address two significant shortcomings in MG2: (a) rimed precipitating ice particles (graupel or hail) are not considered in MG2, and (b) MG2 artificially partitions frozen particles into non-precipitating cloud ice and precipitating snow based on a fixed threshold. In P3, the rimed mass and rimed volume mixing ratios are prognosed separately, allowing a continuous evolution of the rimed fraction and particle density without artificial conversion between ice-phase categories. By addressing these shortcomings, we expect P3 to improve the simulated precipitation rates, cold-phase cloud properties, and aerosol-cloud interactions. Since some major modifications have been made to the P3 in E3SM, we refer to this version as P3E. These changes include (a) replacing the heterogeneous ice nucleation parameterization scheme based on Cooper (1986), which is temperature dependent, with the Classical Nucleation Theory, which depends on both aerosol and temperature (Hoose et al., 2010; Y. Wang et al., 2014); (b) incorporating in situ cirrus formation processes, including homogeneous freezing of sulfate solution droplets and heterogeneous nucleation on coarse-mode dust, which were absent in the original P3, using the approach of Liu and Penner (2005) in P3E; and (c) implementing necessary modifications and adjustments to the scheme originally designed for kilometer-scale models for use in EAMv3 with coarse resolution and large time steps. These adjustments include adding subgrid variability, modifying raindrop evaporation, and introducing additional melting after sedimentation. While the original P3 is designed as a multi-ice category scheme (Milbrandt & Morrison, 2016), the P3E implementation in EAMv3 adopts a single-ice category approach, consistent with the refactored version used in the Simple Cloud-Resolving E3SM Atmosphere Model (SCREAM, Caldwell et al., 2021). Further details on the implementation and performance of P3E can be found in J. Wang et al. (2021) and Shan et al. (2024).

2.1.3.2. A Two-Moment Cloud Microphysics Scheme for Convective Clouds

A two-moment sophisticated microphysics scheme was implemented in EAMv3 in the ZM deep convection scheme, which treats convective cloud microphysics through a simple tuning parameter to convert cloud water to rainwater and neglects the ice-phase microphysical processes that are crucial for deep convection. As such, the latter is not able to properly represent the microphysical processes in different convective regimes or to represent the interaction between aerosols and convective clouds. Therefore, it limits the model's ability to represent the impact of aerosols on climate.

The new cloud microphysics parameterization (Song et al., 2025) is based on the two-moment four-class (cloud water, cloud ice, rain, and snow) convective microphysics scheme of Song and Zhang (2011) by further representing the microphysical processes related to the fifth hydrometeor species, graupel, and the interaction between microphysics and cloud thermodynamics. This scheme explicitly treats the mass mixing ratio and number concentration of the five hydrometeor species and describes several microphysical processes such as autoconversion, self-collection, collection between hydrometeor species, freezing, and sedimentation. In this scheme, raindrops and graupel are treated as precipitating hydrometeors that fall through convective clouds to the surface. Snow, due to its lower density and fall speed, undergoes both detrainment and sedimentation within convective clouds. In addition, aerosols affect convective cloud microphysics through cloud ice and droplet nucleation parameterizations.

Coupling with the stratiform P3E microphysics scheme (which uses a single ice category) is performed as follows: the detrained number concentrations of ice crystals and snow from convection are summed and added to the stratiform ice number concentration as source terms, with the same approach applied to the stratiform ice mass. Detrained cloud droplet mass and number concentrations are also added to the corresponding stratiform prognostic variables. As previously noted, raindrops and graupel are assumed to precipitate out during convection and are not detrained.

2.1.4. Deep Convection

In addition to the new microphysics for convective clouds described above, two additional enhancements were made to the ZM deep convection parameterization. They are described below.

2.1.4.1. Convective Mass Flux Adjustment (MAdj)

EAMv3 revised the calculation of cloud-base mass fluxes in ZM (MAdj) to incorporate the dynamical effects of large-scale vertical motion on convection (Song et al., 2023). With MAdj, convection is enhanced (suppressed) when there is large-scale ascending (descending) motion at the planetary boundary layer top. The coupling of convection with the large-scale circulation significantly improves the simulation of climate variability across multiple scales from the diurnal cycle, convectively coupled equatorial waves, to Madden-Julian oscillations (Song et al., 2023).

2.1.4.2. The Multiscale Coherent Structure Parameterization (MCSP)

EAMv3 includes the Multiscale Coherent Structure Parameterization (MCSP) described in Moncrieff (2004, 2019), Moncrieff and Liu (2006), and Moncrieff et al. (2017) to represent the dynamical and physical effects of organized mesoscale convection, which are often missing in conventional cumulus parameterizations. The current implementation of MCSP only includes its parameterization of mesoscale heating tendencies. The heating induced by organized mesoscale convection is represented in the form of a second baroclinic normal mode with amplitude proportional to the vertically averaged convective heating from the ZM deep convection scheme. Mesoscale heating is added in the stratiform region and evaporative cooling is added beneath, consisting of the

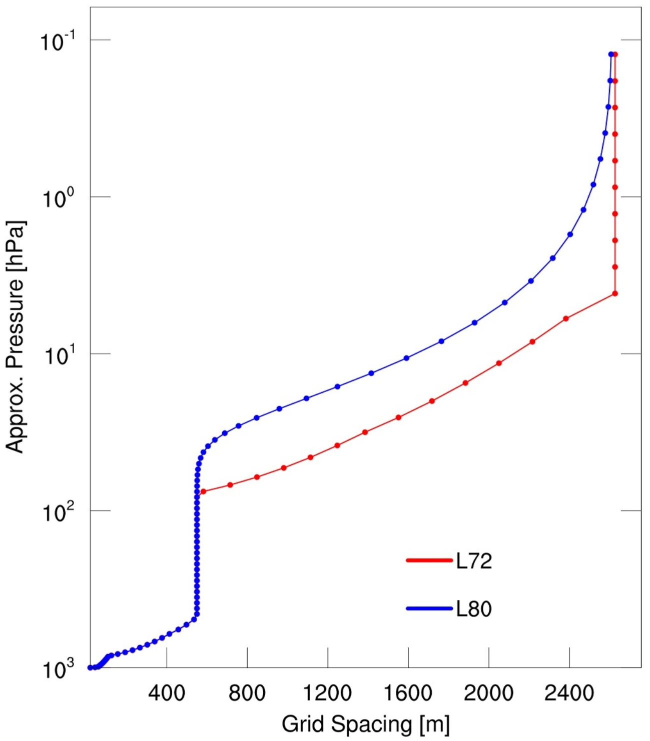

Figure 2. Approximate vertical level thickness versus reference pressure for the EAMv2 72 layer grid (red) and the EAMv3 80 layer grid (blue).

mesoscale ascent and mesoscale cooling descent. C.-C. Chen et al. (2021) found that by adding MCSP, both the representation of large-scale precipitation systems and the modes of variability from tropical waves are improved.

2.1.5. Gustiness Bug Fixes

In addition to the enhancements to the deep convection scheme described above, there are a few important bug fixes made to the convective gustiness scheme (Harrop et al., 2018; H.-Y. Ma et al., 2022; P.-L. Ma et al., 2022; Redelsperger et al., 2000) for EAMv3. These include (a) ensuring that the surface stress is calculated as a vector quantity, rather than using a formula that is only appropriate for mean scalar stress; (b) taking sums of squares, rather than sums of wind magnitudes, to calculate the effects of gustiness on the magnitude of wind, that is correctly implementing Equation 1 of H.-Y. Ma et al. (2022), P.-L. Ma et al. (2022); (c) turning off the land model's parameterization of gustiness from boundary layer turbulence when CLUBB provides it, to avoid double counting; and (d) changing the diagnostic U10 output so that it does not include the gustiness enhancement. Some of the code needed for (a) is not yet implemented in MPAS-SeaIce, so gustiness is applied only over the ocean and land in E3SMv3, but not over sea ice. Initial tests indicate stronger convection over tropical land, which leads to local increases in precipitation and high clouds and decreases in low clouds, weaker latent heat flux over much of the tropical ocean, and higher sensible heat flux over land compared with EAMv2.

2.2. Atmospheric Dynamical Core

EAMv3 contains several minor updates to the EAMv2 dynamical core. A detailed overview of the EAMv2 dynamical core is given by Golaz et al. (2022). In EAMv3, two changes were made to improve the model's treatment of topography. The hyperviscosity operator applied to potential temperature is modified to only act on perturbations from the potential temperature reference state given in Simmons and Jiabin (1991). This has the effect of not dissipating large-scale impacts of topography. Using the same potential temperature reference state and its associated balanced geopotential, the terrain following pressure gradient terms are modified following Herrington et al. (2022). This modification is identical to the original in the continuum but significantly reduces the discretization error over topography. Because of these improvements, EAMv3 uses rougher topography, which is described in Section 2.3.2. Based on the energy spectrum, the EAMv3 topography has approximately 50% less smoothing than the topography data sets used in EAMv2. Finally, EAMv3 changed the hyperviscosity operator used to add horizontal dissipation to the prognostic variables, adopting the tensor-hyperviscosity formulation from Guba et al. (2014). This operator uses tensor coefficients to allow the dissipation to scale with resolution and correctly handle distorted grid cells. It has been used in EAMv1 and EAMv2 regionally refined grids. In EAMv3 it is now also used for quasi-uniform cubed-sphere grids. On cubed-sphere grids, the tuning used by EAMv3 results in 16% less dissipation as compared to EAMv2.

2.3. Model Grid and Boundary Condition

2.3.1. Vertical Resolution

One significant update in the EAMv3 model resolution is the increase from 72 to 80 vertical levels. This change was designed to restructure the stratospheric levels, with refinement concentrated in the lower stratosphere to improve the QBO, while leaving the tropospheric levels and model top unchanged. The distribution of approximate vertical grid spacing for both 72- and 80-level grids is shown in Figure 2. The stratospheric refinement along with surrogate-accelerated multi-objective parameter optimization for parameter retuning has led to a significant improvement in the fidelity of the QBO, especially in terms of the amplitude, as will be discussed in Section 4.1.5.2.

2.3.2. Topography Change

EAMv3 uses a rougher topography than the previous versions of EAM, as mentioned earlier, to allow for a more realistic flow. The rough topography smooths the surface geopotential with six iterations of the Laplace operator, compared to 16 iterations in EAMv1 and EAMv2. In companion, a new pressure gradient correction along with a modification to hyperviscosity to minimize the dissipation of background reference states is employed to minimize unphysical noises that would otherwise be generated when the topography is too rough. The software to compute topography is documented in Lauritzen et al. (2015).

3. Model Configuration, Simulation, and Evaluation Data

3.1. Model Configuration

The standard EAMv3 under the E3SM v3.0.0 code base is evaluated in this study. The model was run on the same ne30pg2 grid as in E3SMv2 and described in Golaz et al. (2022). The ne30pg2 grid uses different grids for dynamics and physics, that is, ne30 dynamics grid (∼110 km) and pg2 physics grid (∼165 km), for improved computational efficiency (Hannah et al., 2021). The vertical resolution used in EAMv3 is increased from 72 layers to 80 layers with the model top unchanged at ∼60 km. The new features described in Section 2 are all turned on in the AMIP simulations analyzed below except for the new MOSAIC nitrate aerosols which are currently not set as the default option for EAMv3 due to its computational burden. The active land component (ELMv3) uses a half-degree rectangular latitude-longitude grid.

3.2. Model Simulation

3.2.1. AMIP Simulations

Three AMIP simulations, which differ in the land, sea-ice, and atmospheric initial conditions taken from the 1 January 1870 states of the E3SMv3 coupled historical simulation members, are used to evaluate EAMv3 model fidelity in the simulation of mean climate and its variability. The simulations span the period from 1870 to 2014, following the AMIP protocol. The active atmosphere and land components are forced by transient prescribed sea surface temperature and sea ice concentration, land use and land cover, greenhouse gas concentrations, aerosols and their chemical precursors, volcanic emissions, top-of-the-atmosphere incoming solar radiative fluxes, linoz data for ozone, and chlorine and bromine concentrations important for stratospheric ozone concentration.

3.2.2. Simulations for Evaluating ERF_{aer}

To evaluate aerosol effective radiative forcing (ERF_{aer}), two simulations were performed using pre-industrial (PI, year = 1850) and present-day (PD, year = 2010) emissions of aerosols and aerosol precursor gases; no eruptive volcanic emissions were included. Emissions of tracers required by the gas-phase chemistry scheme were fixed at their present-day levels. SSTs and sea-ice extents were set at their present-day levels.

Atmospheric variability (e.g., different weather patterns in free-running PD/PI simulations) can obscure the aerosol forcing signal over short time scales (Kooperman et al., 2012). Constraining the simulated large-scale circulation using model outputs or reanalysis data helps to more effectively isolate and identify the aerosol forcing signal (Kooperman et al., 2012; K. Zhang et al., 2014). To achieve a high signal-to-noise ratio in short simulations (for rapid turnaround), we nudge our simulations to the Modern-Era Retrospective analysis for Research and Applications (MERRA) MERRA2 (Gelaro et al., 2017) for 15 months (1 October 2009 to 1 January 2011), with the final 12 months used for analysis, following Mahfouz et al. (2024). We nudge horizontal winds (u, v) at every time step with reanalysis data every 6 hr, with a relaxation timescale of 6 hr. Following C. Zhang et al. (2022), K. Zhang et al. (2022), we apply nudging throughout the atmosphere except in the layers closest to the surface (excluding the lowest 10 layers), to mitigate the influence on the simulated wind-driven surface aerosol emissions relative to the free-running model, though the exclusion of layers does not seem to have an influence on the simulated ERF_{aer} signal in our runs.

Table 2
Summary of Model Evaluation Data

Data	Description	Period	Reference or source	Quantity used in the analysis
GPCP v2.3	Global Precipitation Climatology Project	1979–2017	Adler et al. (2003)	Total precipitation rates
TRMM 3B43v7	The Tropical Rainfall Measuring Mission (TRMM) merged with monthly precipitation from the Global Precipitation Climatology Center (GPCC)'s rain gauge analysis	1998–2013	Huffman et al. (2007)	Total precipitation rates
IMERG	NASA Integrated Multi-satellitE Retrievals for GPM (IMERG) version V06B	2001–2019	Huffman et al. (2019)	Total precipitation rates
ERA5	European Center for Medium Range Weather Forecasts (ECMWF) reanalysis version 5	1979–2014	Hersbach et al. (2020)	Zonal winds, temperature, relative humidity, and 2 m air temperature.
CERES-EBAF v4.1	Clouds and the Earth's Radiant Energy System (CERES) Energy Balanced And Filled (EBAF) observations (Kato et al., 2013)	2001–2018	NASA/LARC/SD/ASDC (2019)	SW and LW cloud radiative effect (SWCRE and LWCRE)
AVHRR	Advanced Very High-Resolution Radiometer (AVHRR) aerosol optical depth (AOD) version 4	1991–1994	Russell et al. (1996), Quaglia et al. (2023)	AOD, SAOD
MACv2	Max Planck Aerosol Climatology version 2 based on AERONET and AeroCom model results	1995–2015	Kinne (2019)	AOD
AERONET	NASA AERONET (AErosol RObotic NETwork) Level 1.5 products based on the Version 3 Direct Sun and Inversion Algorithms	2006–2015	Dubovik et al. (2000)	AOD, AAOD
CASTNET	Clean Air Status and Trends Network	2005–2014	https://www.epa.gov/castnet	Sulfate
EMEP	European Monitoring and Evaluation Programme	2005–2014	https://ebas-data.nilu.no/Default.aspx	Sulfate
EANET	Acid Deposition Monitoring Network in East Asia	2005–2014	https://monitoring.eanet.asia/document/public/index	Sulfate
remote ocean sites	observational data used in AeroCom phase I	1980–2000	Huneeus et al. (2011)	Sulfate
IMPROVE	Interagency Monitoring of Protected Visual Environments	2005–2014	Malm et al. (1994)	BC, OC
OMI + MLS ozone columns	https://acd-ext.gsfc.nasa.gov/Data_services/cloud_slice/new_data.html	2005–2017	Ziemke et al. (2006, 2019)	Tropospheric column ozone (TCO), stratospheric column ozone
NASA Ozone Watch observational data	NASA Ozone Watch	1990–2014	https://ozonewatch.gsfc.nasa.gov/	SH minimum total column ozone, SH maximum ozone hole area

3.3. Evaluation Data

Both satellite and ground-based measurements as well as the fifth-generation European Center for Medium-Range Weather Forecasts (ECMWF) reanalysis data (ERA5) (Hersbach et al., 2020) are used to assess model fidelity. Detailed information about these evaluation data sets is provided in Table 2.

4. Model Fidelity

4.1. Simulated Physical Climate

4.1.1. Global Average Characteristics

Table 3 displays global average quantities of key atmospheric variables in EAMv3 and EAMv2, alongside their observational (reanalysis) counterparts, from the AMIP simulations. Overall, model performance is comparable

Table 3
Annual and Global Average Quantities of Key Atmospheric Variables in EAMv3 and EAMv2 (1985–2014) and Their Observational (Reanalysis) Counterparts

Variable/field (observational reference + units)	Observational reference	EAMv3	EAMv2
2 m temperature over land (ERA5; °C)	8.87	8.59	8.72
Net flux at top of model (CERES-EBAF; W m^{-2})	0.9	2.2	0.2
All sky SW flux at top (CERES-EBAF; W m^{-2})	241.2	240.8	240.3
All sky LW flux at top (CERES-EBAF; W m^{-2})	240.3	238.7	240.2
SWCRE (CERES-EBAF; W m^{-2})	−45.3	−49.2	−47.9
LWRE (CERES-EBAF; W m^{-2})	25.8	24.8	23.7
Surface downward SW flux (CERES-EBAF; W m^{-2})	186.9	186.6	185.0
Surface downward LW flux (CERES-EBAF; W m^{-2})	345.5	345.0	345.4
Water vapor path (ERA5; mm)	24.2	26.0	25.5
Precipitation (GPCP v2.3; mm/d)	2.69	2.99	3.02
Sensible heat flux (ERA5; W m^{-2})	16.7	19.9	20.2
Latent heat flux (ERA5; W m^{-2})	83.1	86.6	87.3
U@200 hPa (ERA5; m/s)	15.7	16.4	16.2
Aerosol optical depth (MACv2; dimensionless)	0.122	0.112	0.159

between EAMv2 and EAMv3. However, EAMv3 slightly reduces the high biases in global mean precipitation and sensible and latent heat fluxes. A significant improvement is observed in Aerosol Optical Depth (AOD), which is notably reduced and aligns more closely with observations. This is mainly due to the improved aerosol wet removal treatments as shown in Shan et al. (2024). This improvement contributes to the improved SW flux at TOA and the surface. The results for the simulated cloud radiative effects are mixed: EAMv3 shows improved LWCRE but a degradation in SWCRE compared to EAMv2. Tuning one of these two often leads to degradation in the other (not shown). The model's mean LWCRE and SWCRE are both too low (Table 3), and tunings that improved the LWCRE tended to increase the amount of high-level clouds, which leads to a more negative SWCRE. The too-strong SWCRE in EAMv3 represents one of the largest degradations seen during model development, mainly due to increased low- and middle-level cloud fractions. These increases in the low- and middle-level cloud fractions are consistent with the inclusion of the convective microphysics scheme (Song et al., 2025). Additionally, EAMv3 experiences a degradation in LW flux at TOA and the surface and has a higher water vapor path compared to both ERA5 and EAMv2. The TOA net energy balance is measured at 2.2 W m^{-2} in EAMv3, larger than 0.2 W m^{-2} in EAMv2 and 0.9 W m^{-2} in CERES-EBAF estimates. This imbalance is partially due to model tuning that focused on achieving a stable coupled pre-industry spinup (piControl). The coupled model performs well in terms of TOA energy balance, maintaining an average imbalance of approximately −0.05 W m^{-2} over the long piControl run. In addition, the historical simulation shows a reasonable 1985–2014 average TOA imbalance of +0.38 W m^{-2}.

Figure 3 shows the Portrait Plots from the Program for Climate Model Diagnosis and Intercomparison (PCMDI) Metrics Package (PMP) (J. Lee et al., 2024) for 27 selected physical quantities, contrasting the E3SM Atmosphere only (AMIP) simulations (rightmost columns) with the 42 AMIP simulations contributed to the Coupled Model Intercomparison Project Phase 6 (CMIP6) (Eyring et al., 2016). The metrics feature the root-mean-square error (RMSE) calculated for seasonal climatology averaged over the 1985–2014 period. The observations for the metrics are from the same climatological period except for the radiative fluxes (i.e., *r** quantities in Figure 3) where available observations only extended from 2001 to 2014.

In general, the performance of EAMv3 is comparable to that of EAMv2. The differences among the three EAMv3 ensemble members are smaller than those between the two versions. EAMv3 slightly reduces biases in surface precipitation (pr), air temperature at 850 hPa (ta-850), and most radiative flux fields at both the surface and top of the atmosphere, while it increases biases in precipitable water (prw), surface upwelling shortwave radiative flux (rsus), surface wind stress (tauu and tauv), and meridional wind at 850 hPa (va-850) compared to EAMv2. Overall, the differences between the two versions are smaller than the inter-model difference among the CMIP6

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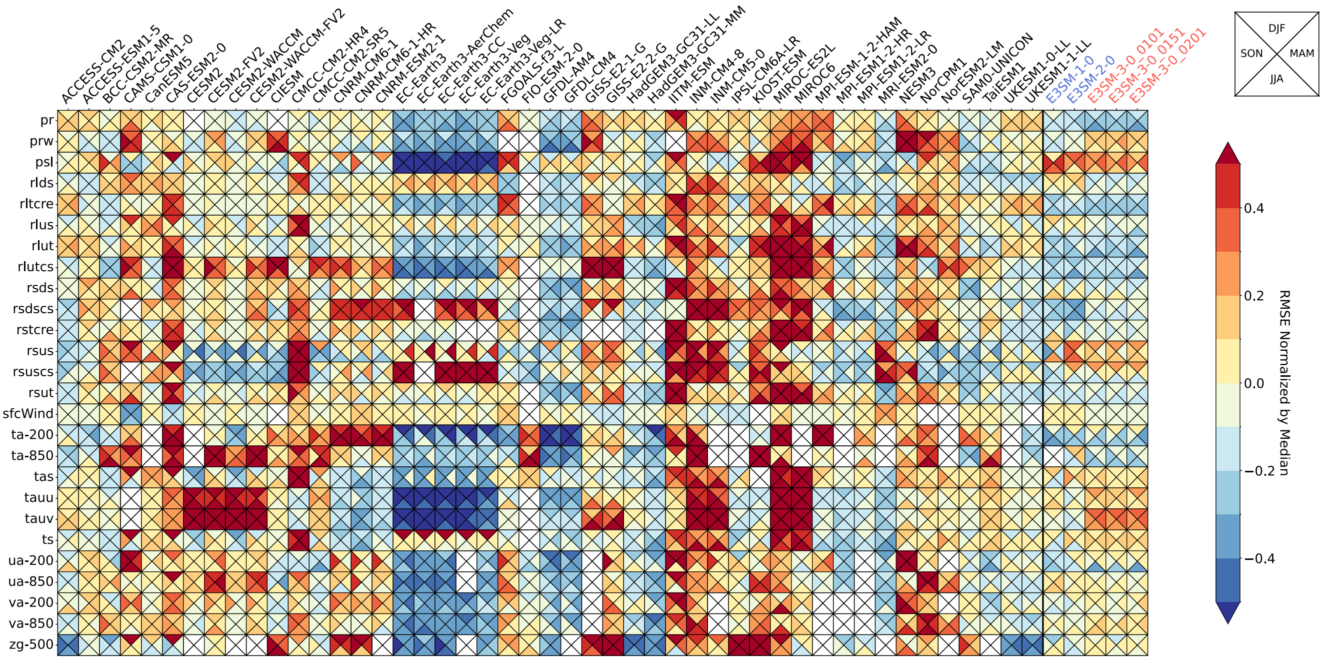

Figure 3. Portrait Plot that compares performance of E3SM simulations and CMIP6 Atmospheric Model Intercomparison Project (AMIP) simulations (first ensemble member) for seasonal climatology of multiple model variables including precipitation (pr), precipitable water (prw), sea level pressure (psl), surface downwelling longwave radiation (rlds), cloud radiative effect on longwave radiation (rltcre), surface upwelling longwave radiation (rlus), top-of-atmosphere (TOA) outgoing longwave radiation (rlut), TOA outgoing clear-sky longwave radiation (rlutcs), surface downwelling shortwave radiation (rsds), surface downwelling clear-sky shortwave radiation (rsdscs), cloud radiative effect on shortwave radiation (rstcre), surface upwelling shortwave radiation (rsus), surface upwelling clear-sky shortwave radiation (rsuscs), TOA outgoing shortwave radiation (rsut), surface wind (sfcWind), air temperature at 200 and 850 hPa (ta-200 and ta-850), near-surface air temperature (tas), surface wind stress (tauu and tauv), surface temperature (ts), wind components at 200 and 850 hPa (ua-200, ua-850, va-200, va-850), and 500 hPa geopotential height (zg-500). All metrics used the 30-year climatology generated with the monthly model output from AMIP simulations during 1985–2014. The Portrait Plot was generated by the PCMDI Metrics Package (Lee et al., 2024) that is implemented in the E3SM zppy workflow. The RMSE is calculated for each season (shown as triangles in each box) over the globe, including both land and ocean, using model and reference data that were interpolated to a common 2.5° × 2.5° grid. The RMSE of each variable is normalized by the median RMSE of all models. A result of 0.2 (−0.2) is indicative of an error that is 20% greater (lesser) than the median RMSE across all models. Further detailed descriptions of the Portrait Plot can be found at Lee et al. (2024). Results from the three AMIP simulations of EAMv3 are shown in the last 3 columns in the figure.

models' AMIP simulations. Specifically, for most atmospheric states at both the surface and upper levels, the E3SM models show competitive performance, with lower biases than most models in the CMIP6. An exception is the mean sea level pressure (psl), which exhibits large biases that persist from EAMv2 to EAMv3, as previously reported (e.g., Golaz et al., 2022).

Further analysis shows that the biases in dynamical variables, such as surface wind stresses at low levels, are primarily due to a much stronger midlatitude jet, especially in the southern hemisphere (SH), and weaker easterlies over the equatorial oceans (not shown). These biases have persisted in EAM since EAMv1 and have also been reported in the CESM family of models. For example, Small et al. (2014; see their Figure 5) and Simpson et al. (2020; see their Section 4.1) found that CESM tends to produce excessively strong winds in extratropical storm track regions, resulting in overestimated time-mean surface wind stress over the ocean.

To address this issue (i.e., the relatively large surface wind stress bias), ongoing work in E3SM involves implementing a suite of orographic gravity wave drag schemes (Xie et al., 2020) in EAMv3. These schemes introduce additional surface drag, including slow-blocking drag, small-scale gravity wave drag, and turbulent-scale orographic form drag. Preliminary results suggest that these schemes effectively reduce surface wind biases, particularly near mountain ranges. The new orographic gravity wave drag schemes are currently being tested in the high-resolution (~0.25°) EAMv3 model. Initial analyses indicate that the strong SH midlatitude jet bias is reduced in the high-resolution configuration. However, further analysis is needed to clearly separate the effects of increased resolution from those of the new topographic wave drag schemes. We plan to address the improvement of these biases in the manuscript on the high-resolution model configuration.

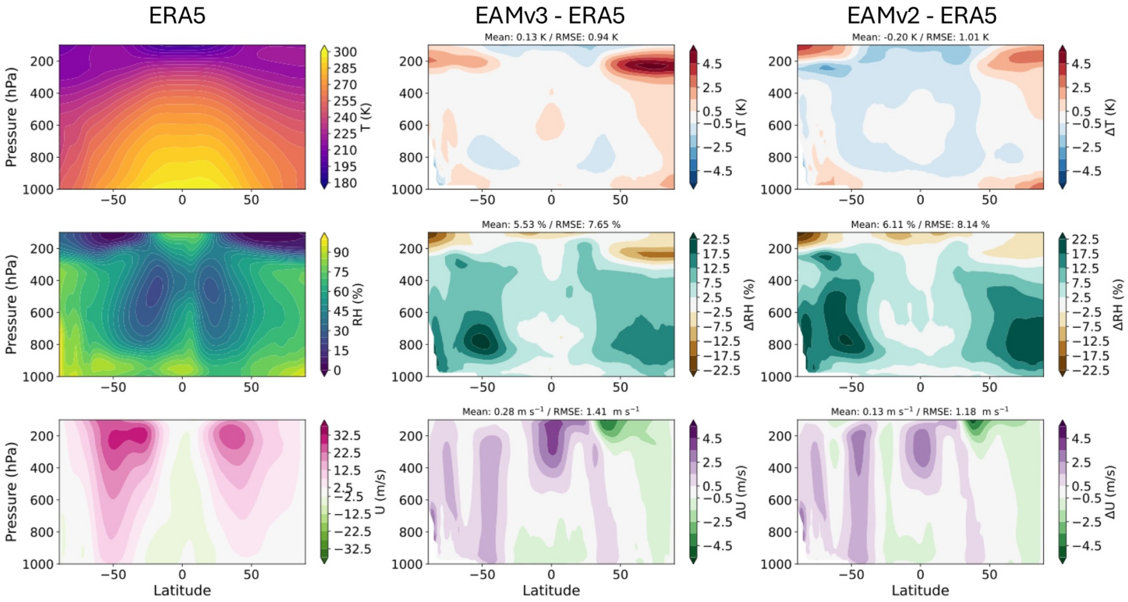

Figure 4. Annual-mean differences in zonally-averaged temperature, relative humidity, and zonal wind profiles in EAMv3 and EAMv2 compared to ERA5. Differences compared to one ensemble member of EAMv3 and EAMv2 (0101) are shown. Zonally averaged ERA5 values are shown as well for reference.

4.1.2. Zonal Mean Meteorological Characteristics

The annually and zonally-averaged profiles of temperature, relative humidity (RH), and zonal wind from ERA5 and the corresponding biases against ERA5 reanalyzes in EAMv3 and EAMv2 are presented in Figure 4. Despite numerous updates in cloud and convection physics, chemistry, and aerosols, the differences in biases in the zonal structure of thermodynamics and zonal wind remain small. Both models exhibit a notable warm bias in the high latitudes and cold bias in the upper troposphere and lower stratosphere in the mid- and low latitudes in both hemispheres. EAMv3 shows a larger warm bias in the polar region and a smaller cold bias in mid- and low-latitudes than EAMv2. Both models exhibit notable overestimation of RH in the extratropics, with the largest bias occurring near the top of the boundary layer. Comparison with other CMIP6 models indicates that the RH bias in EAM models is at the higher end of the CMIP6 range, though still within the ensemble spread (not shown). This bias is partially related to the use of CLUBB in EAM, which tends to increase the moist bias in the extratropics above 850 hPa and reduce the bias in the tropics. However, RH is also influenced by other processes such as deep convection and cloud microphysics. The deficiencies in representing these processes all contribute to the RH bias. Fully understanding the cause of this bias requires further studies to isolate the contributions of individual processes. It is noted that the overestimation of RH in the troposphere seen in EAMv2 is slightly reduced in EAMv3. However, EAMv3 shows a slight degradation in zonal wind simulation, particularly in the upper troposphere over the Tropics.

4.1.3. Latitude/longitude Climate Characteristics

Annual mean SWCRE and LWCRE in CERES-EBAF v4.1 and their difference with EAMv3 and EAMv2 are shown in Figure 5. The global map shows some of the spatial structure of cloud radiative effect differences seen in Table 3. Compared to EAMv2, clouds are more reflective (shortwave cloud forcing more negative) in EAMv3 where differences from EAMv2 are most notable over the tropical trade wind regions, the midlatitudes, and Southern Ocean. It is noted that the large underestimation of SWCRE between 30 and 40S in EAMv2 is significantly reduced in EAMv3. LWCRE is much stronger (more positive in EAMv3) in tropical and subtropical lands and nearby oceans. A notable improvement in EAMv3 is found over the high latitudes where the too-bright bias is reduced and the excessive LWCRE is almost eliminated. As discussed in Terai et al. (2025), these notable changes in both SWCRE and LWCRE, including the reversal of LWCRE error signs in the tropical convection regions, are primarily attributed to the use of P3E for stratiform cloud microphysics, with additional contributions from the updated convective microphysics scheme. The improvement in the Arctic is purely due to the improved mixed-phase clouds from the P3E cloud microphysics (Shan et al., 2024).

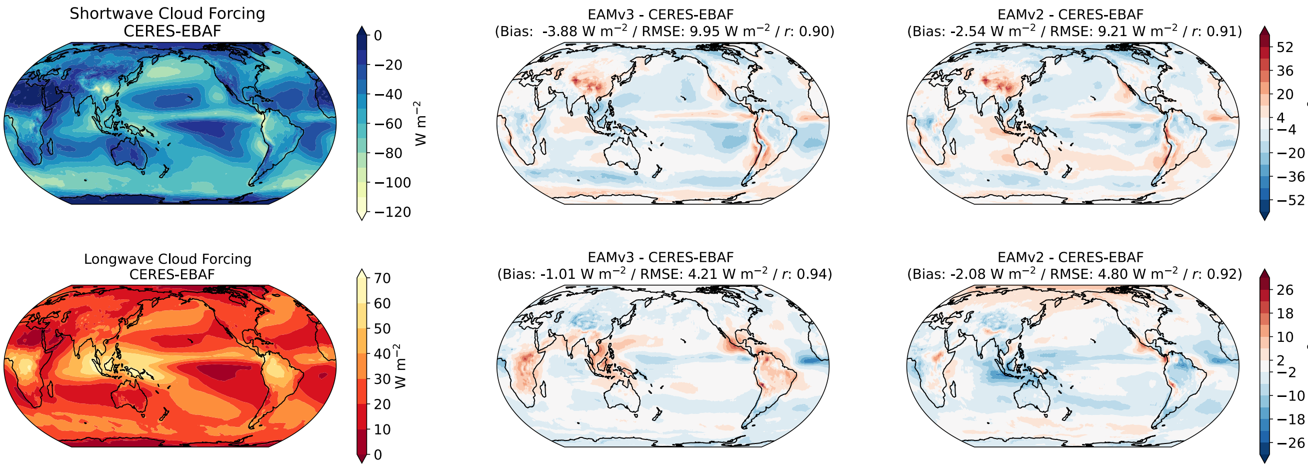

Figure 5. Annual mean shortwave and longwave cloud forcing in CERES-EBAF v4.1 and their difference with EAMv3 and EAMv2. Differences compared to one ensemble member of EAMv3 and EAMv2 (0101) are shown. The bias, RMSE, and correlation coefficient (*r*) (in parentheses) are reported for the panel of the difference between model and observation.

Compared to EAMv2, the annual mean 2-m air temperature bias (Figure 6) is slightly reduced over the Northern Hemisphere (NH) land in EAMv3. EAMv3 still has a too-warm bias over the NH land. Note that warm surface air temperature bias over the NH land is a common model bias shown in many climate models. Earlier studies (e.g., Y. Lin et al., 2017; H. Ma et al., 2018; Morcrette et al., 2018; van Weverberg et al., 2018; C. Zhang et al., 2018) indicated that the bias is likely due to dry bias in precipitation, lack of low clouds, and misrepresentation of land-atmosphere interaction. Therefore, it may not be fully addressed by improvements in atmosphere model alone.

Figure 7 shows precipitation in the annual mean (ANN) and DJF and JJA seasons in the GPCP v2.3 data set (Adler et al., 2003) and the differences between EAMv3 and EAMv2 and GPCP. As with other fields, many of the structural biases in EAMv2 carry over to EAMv3, but the spatial pattern of precipitation is generally improved in EAMv3 across seasons. Examples include reductions in the excessive precipitation simulated over the Eastern Pacific, mountainous regions, and the northwestern Tropical Pacific. There is also an indication of improved precipitation in JJA over the continental US, which is a bias that's shared by many models and has existed in all previous versions of EAM. However, precipitation over the storm tracks in the NH is considerably underestimated in DJF in EAMv3.

It is worth noting that, although the changes in total precipitation are small between these two versions, the fraction of convective precipitation has increased significantly over tropical deep convection regions in EAMv3 compared to EAMv2 (Figure 8). On average, between 30°S and 30°N, convective precipitation accounts for 66% in EAMv3, increased from 59% in EAMv2. This increase is primarily because the MAdj scheme ties deep

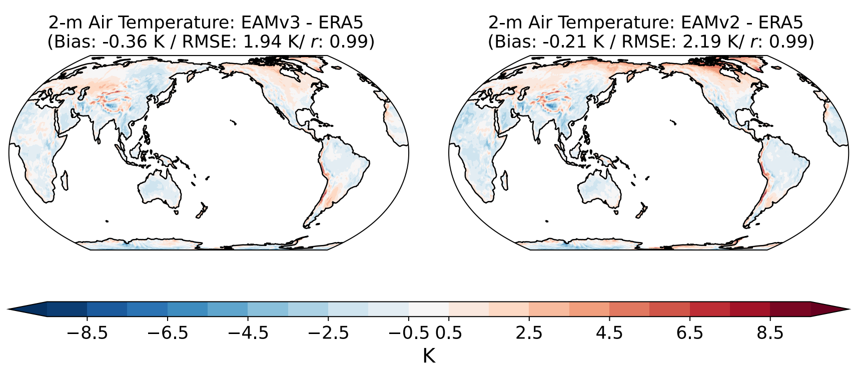

Figure 6. Annual-mean difference in 2-m air temperature between EAMv3 and EAMv2 and ERA5 reanalysis over land. Differences compared to one ensemble member of EAMv3 and EAMv2 are shown. The bias, RMSE, and correlation coefficient (*r*) (in parentheses) are reported for the panel of the difference between model and ERA5 reanalysis.

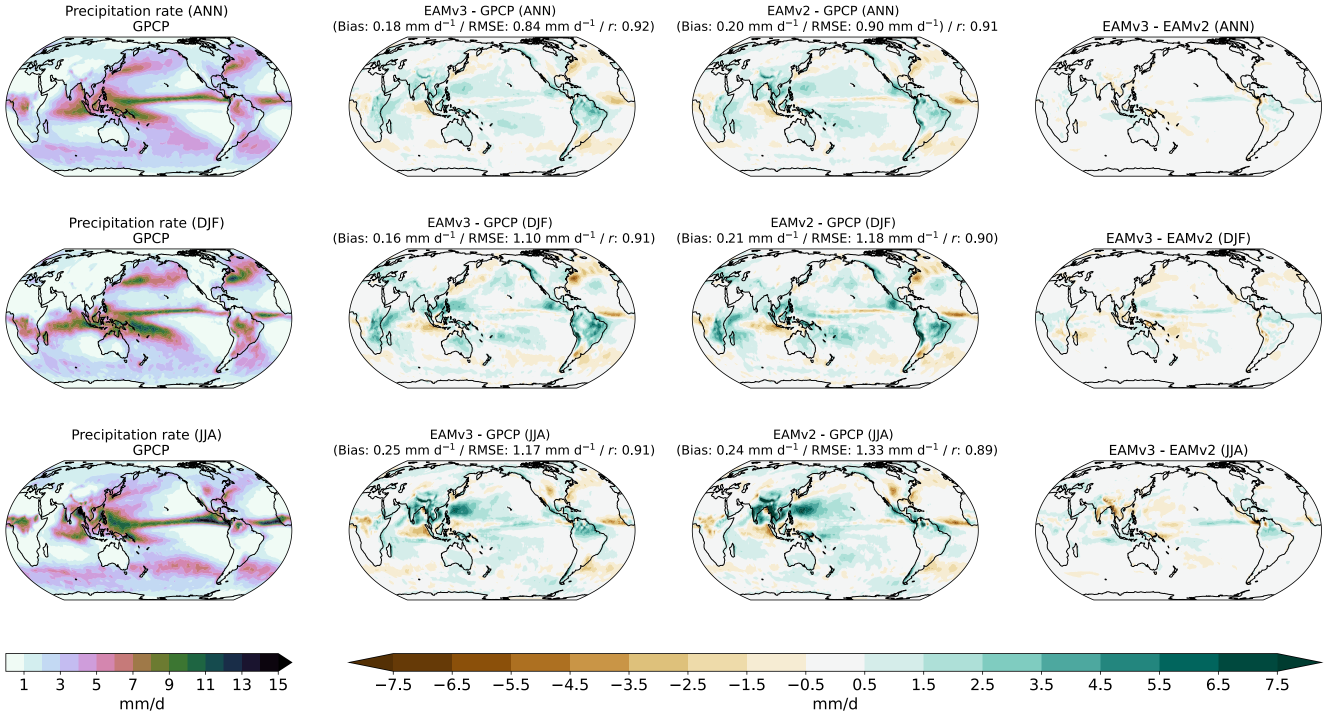

Figure 7. Precipitation in the annual mean (ANN) and DJF and JJA seasons in GPCP v3.2 data set and the differences between EAMv3 and EAMv2 and the satellite estimates and difference between EAMv3 and EAMv2. Differences compared to one ensemble member of EAMv3 and EAMv2 are shown. The bias, RMSE, and correlation coefficient (*r*) (in parentheses) are reported for the panel of the difference between model and observation.

convection more closely to the vertical velocity at the top of the boundary layer, enhancing convective activity during periods of upward motion (Song et al., 2023). As a result, EAMv3 is likely underestimates the fraction of stratiform precipitation (34%) compared to Tropical Rainfall Measuring Mission (TRMM) observations (Kummerow et al., 2000), which suggests the fraction of stratiform precipitation is approximately 35%–55% over most low-latitude regions (Dai, 2006). However, we caution that direct comparison between model output and TRMM is complicated by differences in the definitions of large-scale precipitation and by uncertainties in the TRMM data. Employing a TRMM simulator could improve the comparability between model and observational data, but this is beyond the scope of the current study.

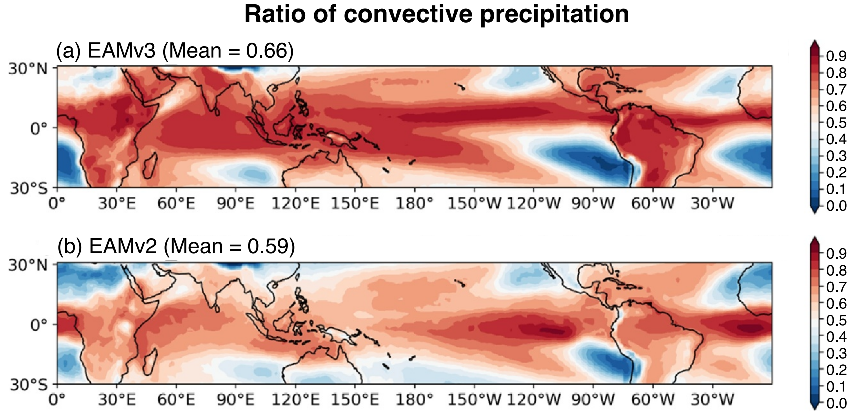

Figure 8. Annual mean convective precipitation ratio, calculated as the convective precipitation rate divided by the total precipitation rate. Results are shown for (a) EAMv3 and (b) EAMv2, with mean values indicated in parentheses.

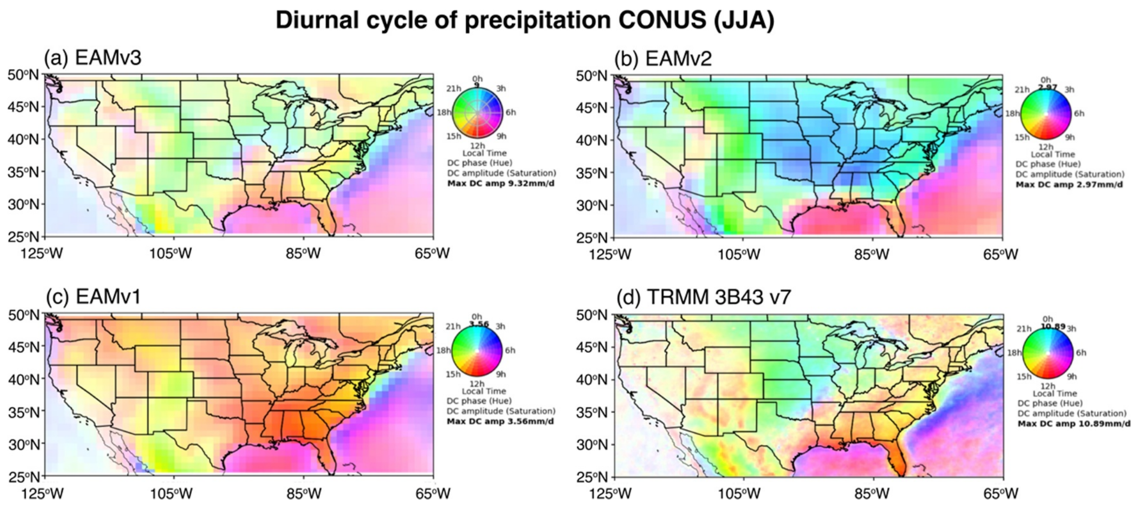

Figure 9. Diurnal cycle of precipitation in EAMv3, EAMv2, EAMv1, and the Tropical Rainfall Measuring Mission (TRMM) 3B43v7 data set over the continental US during the JJA season. The diurnal cycle from one ensemble member is shown in all EAM cases. The averaging period is 1985–2014 for the models and 1998–2013 for TRMM. Diurnal phase is indicated by color, and diurnal amplitude by color density. The color scale is normalized to the maximum diurnal cycle amplitude ("Max DC amp"), which varies among EAM models and TRMM, as shown in the color dial.

4.1.4. Diurnal Cycle of Precipitation

As indicated in Golaz et al. (2022), the simulation of the diurnal cycle of precipitation in EAMv2 has been largely improved across most regions, due to a new convective triggering function implemented in ZM (Xie et al., 2019). However, EAMv2 fails to capture the observed late afternoon peak over the eastern and southeastern US, instead producing a late evening peak in these regions. Figures 9 and 10 present the diurnal cycle of precipitation in EAMv3, EAMv2, EAMv1, and the TRMM 3B43v7 data set over the contiguous US (CONUS) during the JJA season and the Tropics (20N–20S) in annual mean, respectively. Over CONUS, EAMv3 better simulates the timing of the diurnal peak in most regions, particularly in the eastern and southeastern US, where the late afternoon peak is more accurately represented (except along the East coast) compared to EAMv2. However, EAMv3 degrades its skill to capture the nocturnal precipitation observed over the central Great Plains, where EAMv3 shows a precipitation peak in the early evening. A further investigation indicates that the changes in convective cloud microphysics and cloud-base mass fluxes made in ZM are the major cause of this degradation (Song et al., 2023).

In the Tropics, the observations typically reveal late afternoon or early evening peaks in central tropical land areas, a midnight peak near the coast, and an early morning peak around 0500–0700 LST over the oceans. Both EAMv2 and EAMv3 well capture the early morning peak over the oceans; however, EAMv3 largely improves the timing of the diurnal peak over tropical lands, whereas EAMv2 fails to capture the late afternoon or early evening peaks in central tropical land areas, instead showing a peak time near midnight. Despite improvements from EAMv2 to EAMv3, the diurnal amplitude over the Tropics is still substantially underestimated in EAMv3 compared to TRMM.

It is important to note that the diurnal cycle of precipitation over land is more accurately represented in EAMv2 and EAMv3 than in EAMv1, where the precipitation peak consistently occurs around local noon over CONUS and the tropical lands, which is a common error in most CMIP models (Q. Tang et al., 2021; S. Tang et al., 2022; Tao et al., 2023, 2024). This is likely due to problems with parameterized convective clouds as global kilometer scale models have demonstrated a better skill in simulating diurnal cycle of precipitation (H.-Y. Ma et al., 2022; P.-L. Ma et al., 2022, Z. Feng et al., 2023).

4.1.5. Tropical Variability

4.1.5.1. Intraseasonal Variability

Despite considerable progress, accurately simulating the space-time distribution of tropical subseasonal convection remains a steep challenge for global climate models (Ahn et al., 2020; G. Chen et al., 2022). This

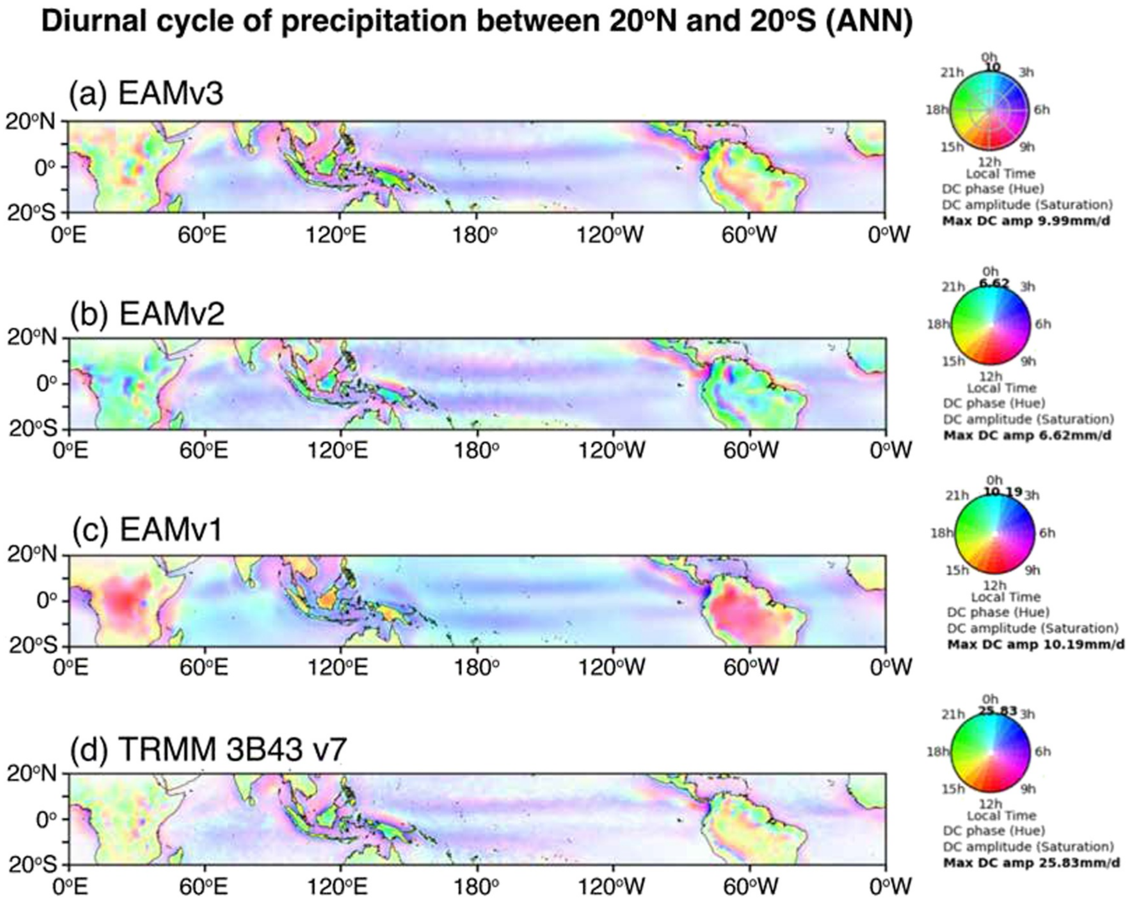

Figure 10. Annual-mean diurnal cycle of precipitation in EAMv3, EAMv2, EAMv1, and the Tropical Rainfall Measuring Mission (TRMM) 3B43v7 data set over the Tropics (20N–20S). The diurnal cycle from one ensemble member is shown in all EAM cases. The averaging period is 1985–2014 for the models and 1998–2013 for TRMM. Diurnal phase is indicated by color, and diurnal amplitude by color density. The color scale is normalized to the maximum diurnal cycle amplitude ("Max DC amp"), which varies among EAM models and TRMM, as shown in the color dial.

challenge arises in part because (a) organized convection manifests across a wide range of space-time scales, (b) convective systems are sensitive to small-scale processes that must be parameterized in current climate models, and (c) a complete understanding of key mechanisms has yet to be achieved (Jiang et al., 2020). A standard approach to succinctly examine variability associated with subseasonal tropical convective disturbances involves space-time spectral decomposition of precipitation in zonal wavenumber-frequency space (Wheeler & Kiladis, 1999). Here, precipitation is examined given its direct ties to the condensational heating that drives anomalous circulations both within and outside the tropics. For Figures 11 and 12, all data sources have been interpolated to a $1^\circ \times 1^\circ$ latitude-longitude grid. The resulting spectra for the component of precipitation that is symmetric about the equator indicate the total power (Figure 11, top row) in E3SMv2 and E3SMv3 is underestimated at all zonal wavenumbers and frequencies compared to precipitation estimates from the NASA Integrated Multi-satellitE Retrievals for GPM (IMERG) version V06 B product (Huffman et al., 2019). However, the shape of the power distribution in E3SMv3 (Figure 11c) appears more similar to observations (Figure 11a) compared to E3SMv2 (Figure 11b). Dividing each total power spectrum by its associated smoothed background spectrum reveals prominent disturbance types (Figure 11, bottom row) that stand above the signal-to-noise ratio. Compared to E3SMv2 (Figure 11e), E3SMv3 (Figure 11f) exhibits significantly more realistic Kelvin waves ("Kelvin"), slightly stronger equatorial Rossby waves ("ER"), and a more robust MJO. In both E3SM versions, the local maximum in spectral power associated with westward inertia-gravity waves ("WIG") remains underestimated.

Spectra for the component of precipitation that is asymmetric about the equator are displayed in Figure 12. Like the symmetric spectra (Figure 11), the shape of the power distribution is improved in E3SMv3 (Figure 12c) but

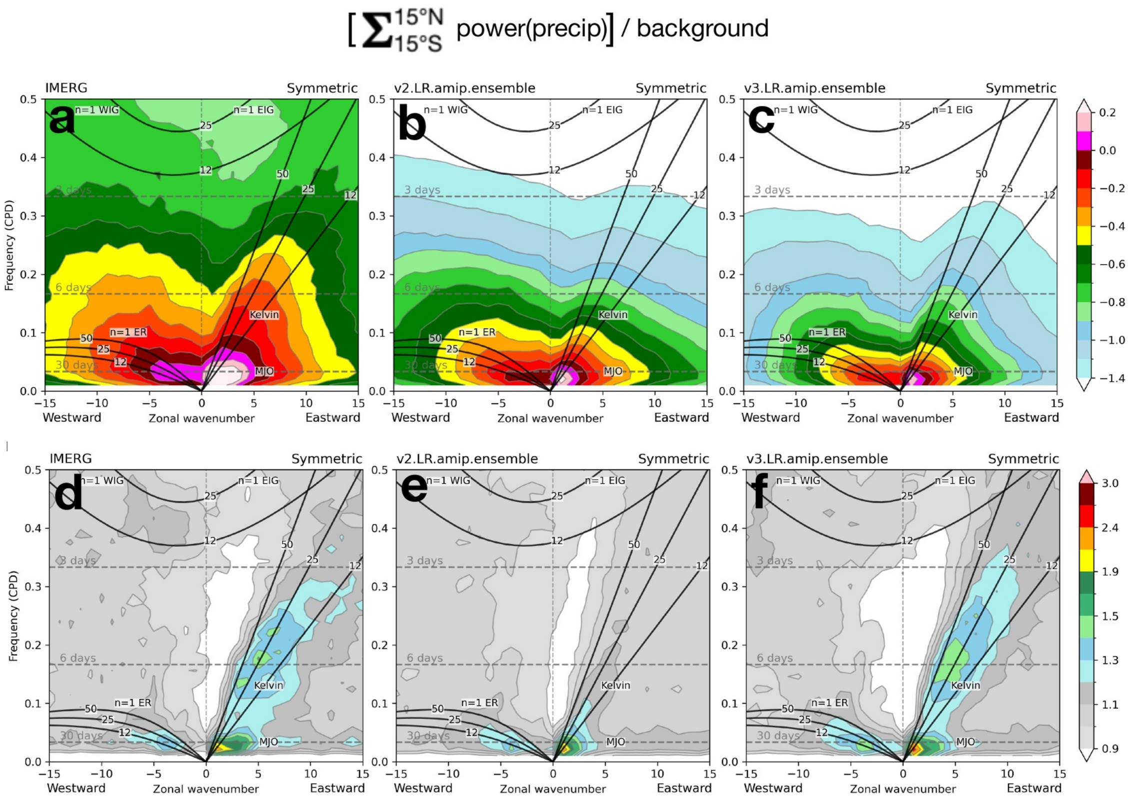

Figure 11. Power spectra in zonal wavenumber-frequency space of the component of precipitation that is symmetric about the Equator, using the methods of Wheeler and Kiladis (1999). Non-normalized power values are shown on the top row, with power values normalized by a smoothed background spectra on the bottom row, for (a, d) IMERG observation-based precipitation estimates, (b, e) E3SMv2, and (c, f) E3SMv3. Panels showing E3SMv2 and E3SMv3 results represent averaged spectra taken from Atmospheric Model Intercomparison Project simulations that were initialized at years 101, 151, and 201 of the respective piControl simulations. Listed disturbance types include: Kelvin waves ("Kelvin"), equatorial Rossby waves ("$n = 1$ ER"), the Madden-Julian oscillation ("MJO"), and westward ($n = 1$ "WIG") and eastward ("$n = 1$ EIG") inertia-gravity waves, where n is the meridional mode number.

the power amplitudes are lower than in E3SMv2 (Figure 12b) and much lower than for IMERG precipitation estimates (Figure 12a). However, signal-to-noise power ratios in E3SMv3 (Figure 12f) are significantly improved for mixed Rossby-gravity waves ("MRG") and eastward inertia-gravity waves ("EIG") compared to E3SMv2 (Figure 12e).

Overall, the precipitation spectra indicate that E3SMv3 produces weaker total power but a stronger signal-to-noise ratio for key intraseasonal convectively coupled equatorial waves, notably, Kelvin and mixed Rossby-gravity waves and the MJO, suggesting that convective organization and scale selection is more realistic in E3SMv3. The improved organization arises from multiple updates to model physics as described earlier, including the more sophisticated microphysics scheme applied to convective clouds (Section 2.1.3.2), the revised convective mass flux adjustment calculation (Song et al., 2023; Section 2.1.4.1), and the implementation of MCSP (C.-C. Chen et al., 2021; Moncrieff & Liu, 2006; Moncrieff et al., 2017). A more complete assessment of intraseasonal convectively coupled equatorial waves will appear in a forthcoming paper.

4.1.5.2. QBO

A prominent feature of the tropical lower stratosphere is the QBO of the zonal mean wind, which can influence variability in both the stratosphere and troposphere. Despite decades of observations and investigation, many models cannot simulate a realistic QBO (e.g., Richter et al., 2020), which is often related to inadequate vertical resolution that prevents the representation of waves with small vertical wavelengths (500–700 m and their wave-mean flow interactions (Richter et al., 2014).

During EAMv3 development the 72-level vertical grid (L72) was modified to have eight additional levels (L80) targeted at lower stratospheric refinement (see Section 2.3.1) in order to improve the representation of the QBO by

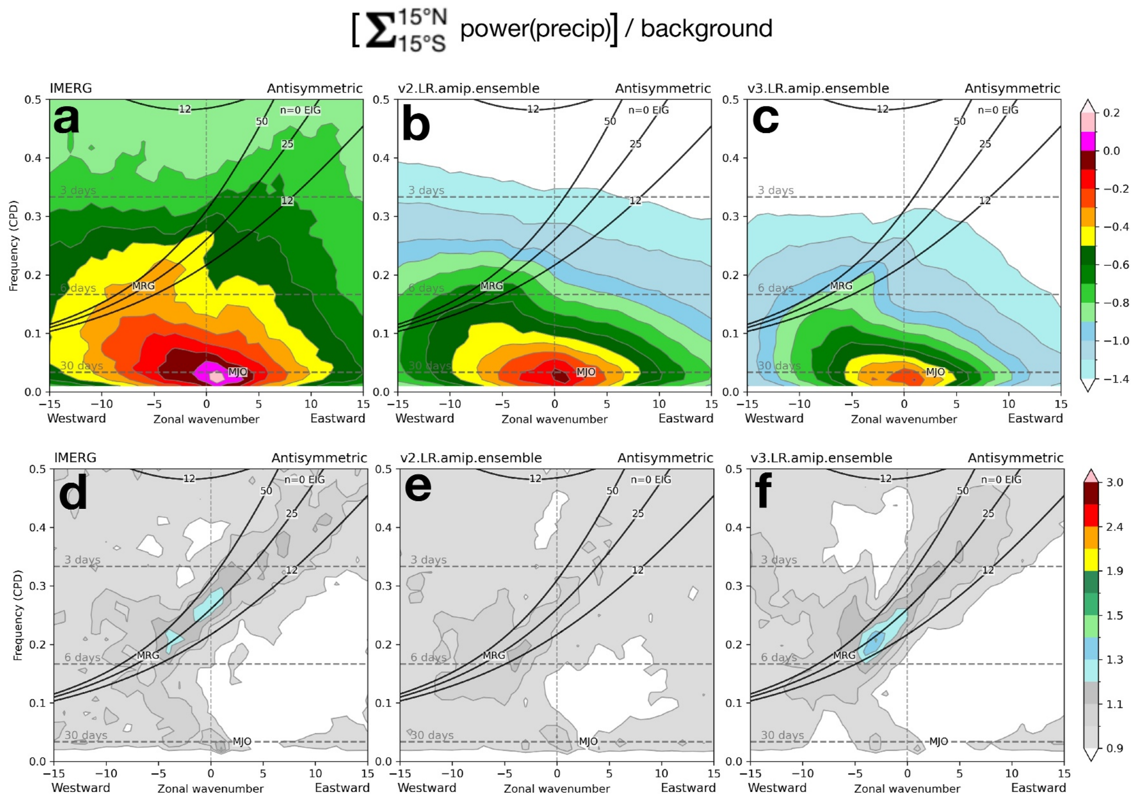

Figure 12. As in Figure 11, but for the component of precipitation that is asymmetric about the Equator. Listed disturbance types include mixed Rossby-gravity waves ("MRG"), the Madden-Julian oscillation ("MJO"), and eastward inertia-gravity waves ("$n = 0$ EIG"), where n is the meridional mode number.

better simulating vertical propagation/dissipation of planetary waves with smaller vertical wavelengths. As detailed by Yu et al. (2025), the L72 grid was too coarse to represent the vertically propagating waves that break in the shear zones of the QBO and drive it downward. Thus, the adoption of L80 improves the QBO by enhancing the resolved wave forcing. The parameterized gravity wave drag is not directly affected by the change in the vertical grid, but still plays an important role in driving the QBO once the forcing by resolved wave is sufficiently large. Experiments with additional refinement showed promise for further improvement of the representation of vertically propagating waves and their impacts on the stratospheric momentum, but the addition of 8 levels was determined to be a reasonable tradeoff between QBO fidelity and computational performance (see Section 5).

The adoption of L80 in EAMv3 also required retuning of the convectively generated gravity wave drag parameterization (see Beres et al., 2004), which was accomplished through a surrogate-accelerated parameter optimization. This targeted three parameters of the convective gravity wave scheme (convective heating conversion factor, gravity wave drag efficiency factor, and heating depth scaling factor) in order to optimize simple metrics of the QBO period and amplitude via a probabilistic surrogate model (see Damiano et al., 2025). Although this effort ultimately left the convective heating rate conversion factor (gw_convect_hcf = 10.0) and convective gravity wave efficiency factor (effgw_beres = 0.35) unchanged from the EAMv2 default values, a convective heating depth scaling factor implicitly set to 1.0 in EAMv2 was added as a new namelist parameter in EAMv3 and changed to 0.5 based on the optimization results. This heating depth scaling factor scales the depth of convective heating diagnosed by the convective gravity wave scheme, which influences the wavelengths of the wave packet. However, the direct influence of this change is difficult to characterize, so we only focused on the impact of parameter changes on the emergent QBO characteristics. Additional optimal parameter sets were identified by the surrogate optimization workflow yielded similarly improved QBO period and amplitude, but did not address the remaining biases in the QBO representation as discussed below.

Figure 13 shows a pressure versus time Hovmoller diagram of stratospheric zonal wind averaged over the 5°S–5°N equatorial band for ERA5, EAMv2, and EAMv3. EAMv3 exhibits an oscillation comparable to

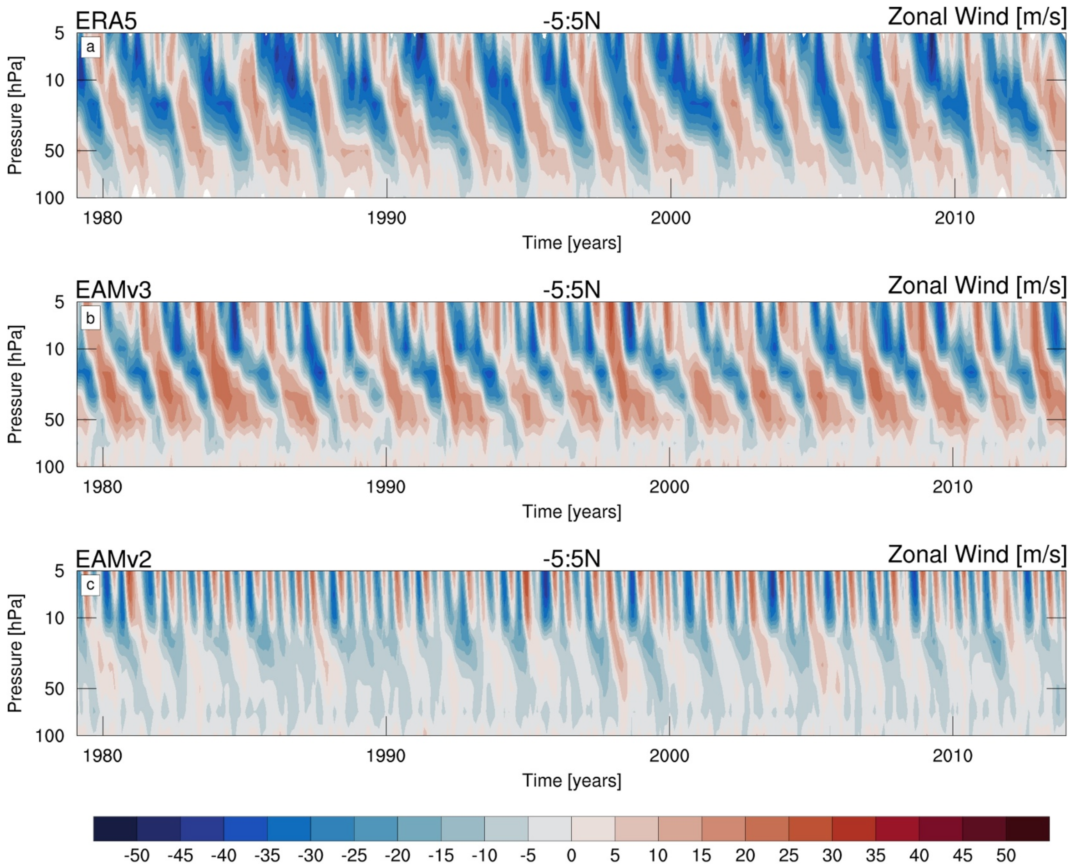

Figure 13. Pressure versus time Hovmoller diagram of zonal wind averaged over a narrow equatorial band (5°S–5°N) for (top) ERA5, (middle) EAMv3, and (bottom) EAMv2.

ERA5 with downward propagation of easterly and westerly wind regimes of similar amplitude to observations, a very significant improvement compared to EAMv2 which shows a much weaker signal dominated by downward propagating easterlies (Golaz et al., 2022). The large improvement in QBO amplitude is further elucidated by Figure 14 showing the vertical profile of QBO amplitude derived from the power spectrum of equatorial zonal wind with periods of 20–40 months. The QBO amplitude for this period band in E3SMv3 is very close to observations between 10 and 100 hPa, with a small deficiency in lower stratospheric amplitude still remaining. One notable deficiency in EAMv3 is that the westerly regimes are stronger than the easterly regimes, which is the opposite of what is observed.

Figure 15 shows the wavelet power spectrum (Torrence & Compo, 1998) of 20 hPa zonal wind over the equatorial region (5°S–5°N) using data from ERA5 (black) and AMIP ensemble simulations with EAMv2 (dashed) and EAMv3 (solid). This highlights that while the EAMv3 QBO amplitude is comparable to ERA5, the period is slightly too short on average.

4.2. Simulated Atmospheric Composition

4.2.1. Ozone

EAMv3 substantially enhances the atmospheric chemistry capabilities relative to the stratosphere-only ozone chemistry in EAMv2. In this section, we focus on the ozone results. More details about EAMv3 chemistry and other gases are documented in Q. Tang et al. (2025).

In the stratosphere, the zonal mean stratospheric column ozone (SCO) (Figure 16a) is well reproduced compared to the satellite data set (Ziemke et al., 2006, 2019) based on the Ozone Monitoring Instrument (OMI) and Microwave Limb Sounder (MLS) measurements (Figure 16b). The SCO has smaller values in the tropics than in the

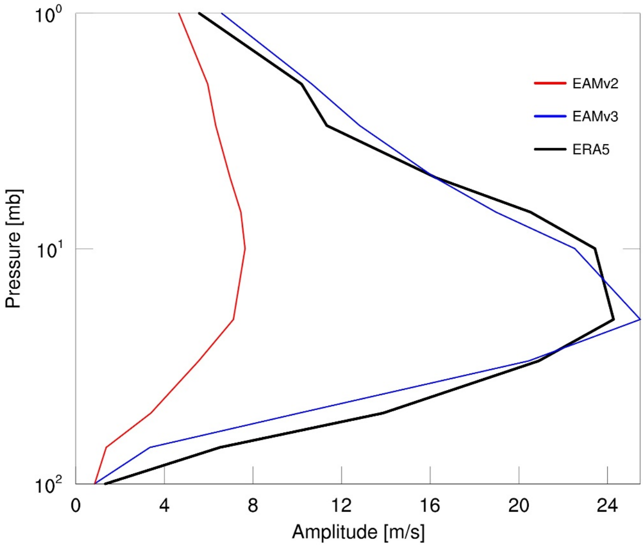

Figure 14. QBO amplitude profile derived from the power spectra of the equatorial mean (5°S–5°N) zonal wind calculated for periods between 20 and 40 months for ERA5 (black), EAMv2 (red), and EAMv3 (blue).

extra-tropics and peaks during winter-spring time in each hemisphere. Both EAMv3 (Figures 16c and 16d) and EAMv2 (Figure 16d) show small biases (up to ±10% with most places of a few percent) relative to observations. The SCO biases are clearly reduced in SH high latitudes due to the better representation of the Antarctic ozone hole in EAMv3 (Figures 16e–16h). The SCO changes in the tropics are because of the more sophisticated Linoz v3 scheme than Linoz v2 and the changes in dynamics (e.g., refer to the QBO section) between the two EAM versions.

The improved Antarctic ozone hole in EAMv3 is mainly because the minimum SH total column ozone is systematically reduced by about 30 Dobson units (DU) (Figure 16e). This helps exceed the 220 DU ozone hole threshold used in the NASA Ozone Watch observational data (https://ozonewatch.gsfc.nasa.gov/), hence leading to better agreement in the ozone hole area (Figure 16f). Despite the EAMv3 improvement over EAMv2, the ozone hole magnitudes are still not as strong as observed. Furthermore, the daily mean climatology results (Figures 16g and 16h) reveal similar ozone hole onset and recovery timings, and interannual variabilities (i.e., standard deviations in shadings) among models and observations. Since the polar stratospheric cloud (PSC) chemistry (Cariolle et al., 1990) has not changed in EAMv3, the better results of the ozone hole are expected to stem primarily from changes in the polar stratospheric dynamics and polar vortex, and the slightly higher PSC chemistry temperature threshold from 197.5 K in EAMv2 (Q. Tang et al., 2021) to 198 K in EAMv3.

In the troposphere, the chemUCI chemistry simulates reasonable ozone compared to observations. Figure 17a shows the global annual mean tropospheric ozone burden from EAMv3 and CMIP6 models over the historical period. EAMv3 is within the multi-model range and simulates very similar time series to CESM and MRI until the 1960s. After the 1960s, EAM3 tropospheric ozone increases at a slightly slower rate than CESM and MRI but is mostly within the one standard deviation range of monthly variabilities in each year. The climatological geographic pattern of annual mean tropospheric column ozone (TCO) of the last three decades (1985–2014) of the simulation is compared with the OMI + MLS observation in Figures 17b–17d. The satellite observations are for 60°S–60°N, where more reliable measurements are available. EAMv3 generally captures the observed TCO patterns with lower values over the Pacific Ocean and higher values in the subtropics and tropical Atlantic Ocean and Africa. The model global mean is 2.7 DU (8.6%) lower than the observation with regional positive biases over tropical Africa, South America, and the Maritime Continent. The high biases appear to result from excessive O3 production linked to deep convection (with lightning NO_x) and/or biomass burning (with O3 precursors) (Q. Tang et al., 2025). It is worth highlighting that the improved atmospheric chemistry in EAMv3 simulates more realistic TCO time evolution primarily driven by emissions, which is missing in EAMv2 with the ozone e-folding decay toward the fixed 30 ppb concentration in the near-surface layers. Hence, the EAMv3 chemistry improves the representation of the time-evolving effective radiative forcing of greenhouse gases and contributes to the improved time series of the surface air temperature anomalies of EAMv3.

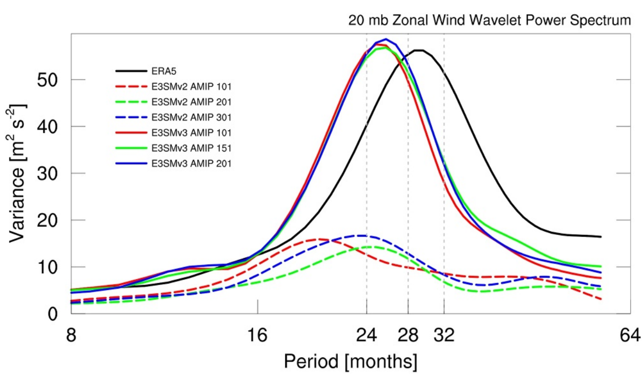

Figure 15. Wavelet power spectra of 20 mb zonal mean zonal wind averaged over the equatorial region (5°S–5°N) and a 35-year time period covering 1979–2014 for ERA5 and Atmospheric Model Intercomparison Project simulations with EAMv2 (dashed) and EAMv3 (solid). The wavelet analysis was performed with the Morlet wavelet of degree 6.

4.2.2. Aerosols

As described in Section 2.1.2, EAMv3 includes several improvements to aerosol-related processes or new aerosol features, compared to previous model versions. Here we use available aerosol measurements from surface monitoring networks, ground-based and satellite remote sensing products to assess the fidelity of aerosol simulations in EAMv3, in the context of comparison with EAMv2. Note that measurement uncertainties in the observational data can arise from many sources related to instrumentation, sampling strategy, spatiotemporal variability and data processing, which may also

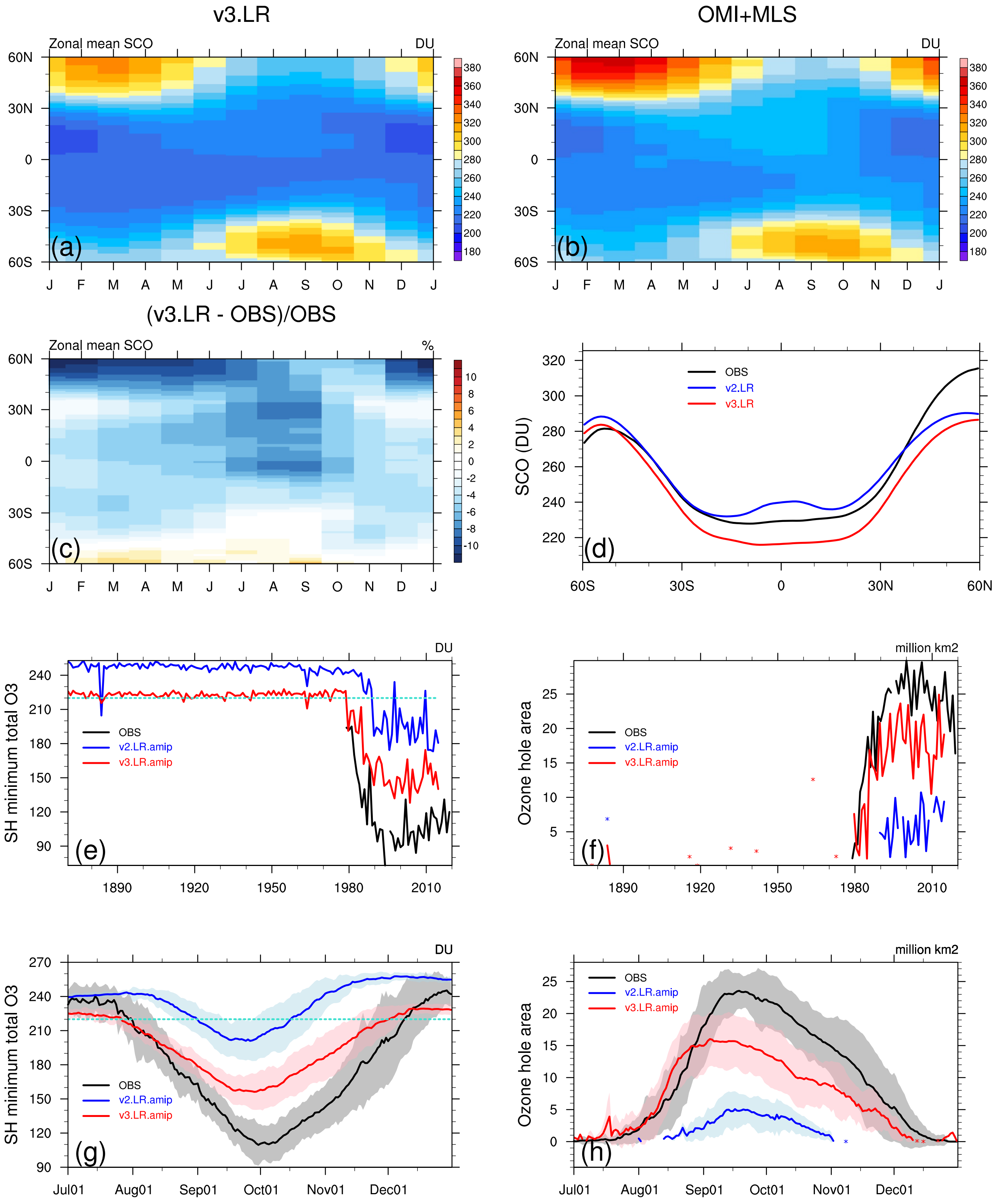
v3.LR
OMI+MLS
Zonal mean SCO
DU
(v3.LR - OBS)/OBS
%
SCO (DU)
OBS
v2.LR
v3.LR
SH minimum total O3
Ozone hole area
million km2
v2.LR.amip
v3.LR.amip

Figure 16.

contribute to the differences between model results and aerosol measurements. Therefore, the aerosol measurements are used here more like a reference than a ground truth to compare changes between EAMv3 and EAMv2.

Figure 18a shows the spatial distribution of the model-calculated annual mean AOD for 2000–2014. High AODs are seen over major aerosol source regions dominated by anthropogenic emissions, such as East and Southeast Asia, and desert dust regions like the Middle East and the Sahara. Moderate AODs are observed in biomass burning regions, including Central Africa, South America, as well as parts of Europe and the eastern United States. The modeled geographical distribution of AOD is largely in agreement with the 10-year AERONET surface observations averaged over 2006–2015, with a correlation coefficient of 0.76, as shown in Figure 18c. The multi-site mean AOD simulated by EAMv3 is 0.124, which is lower than the AERONET mean of 0.211. The model underestimation is more pronounced in regions with high AOD (AERONET AOD greater than 0.1), where the model to measurement ratio is around 1:2, possibly related to the underrepresented anthropogenic emissions, for example, in East and Southeast Asia. This discrepancy may also be due to the model's AOD predictions representing grid box means with a grid size of ~100 km, whereas AERONET data are point-based and likely more sensitive to peak events. On the global mean basis (Table 3), the model-predicted annual AOD is 0.112, which is similar to that derived from the aerosol climatology composite data set (MACv2; Kinne, 2019) at 0.122 and substantially improved from 0.159 in EAMv2.

In addition to AOD, absorption aerosol optical depth (AAOD) is also an important metric for aerosol simulations, especially for the calculation of aerosol direct radiative effect. The comparison with the AERONET data (Figures 18b and 18d) shows that the E3SMv3 simulated grid-mean aerosol absorption agrees well with the point measurements, with a correlation coefficient of 0.84 and RMSE of 0.006. This suggests that the lower AOD in E3SMv3 than in the AERONET data and E3SMv2 is more likely attributed to the uncertainties in representing the scattering aerosol species such as sulfate and organics that lead to an underestimation of aerosol loading.

Figure 19 shows the comparison of annual mean surface concentrations of sulfate, black carbon (BC) and OC simulated in EAMv2 and EAMv3 with observations made at various surface monitoring sites, including the Interagency Monitoring of Protected Visual Environments (IMPROVE) and the Clean Air Status and Trends Network (CASTNET) over US, the European Monitoring and Evaluation Programme over Europe, and the Acid Deposition Monitoring Network in East Asia (EANET) over East Asia. While the spatial correlation between EAMv3/v2 and observations is strong, sulfate aerosol in EAMv3 is generally lower than in EAMv2, underestimating the surface measurements over all regions, especially where the measured concentrations are relatively high (greater than 1 μg m^{-3}), presumably near emission sources, and the ratio of simulated to measured SO_4 concentrations moves away from 1:1 to 1:2. However, both BC and OC concentrations simulated in EAMv3 are slightly closer to the site measurements than in EAMv2. The better agreement of EAMv3 with IMPROVE measurements in OC is likely due to more SOA production in the new VBS scheme.

According to the evaluation of AOD and AAOD results in Figure 18, the EAMv3 performance in simulating column-integrated aerosol properties is likely similar to that at the surface. Such potential aerosol low biases warrant further investigation of the representation of sulfate and organics removal processes as well as their coupling with atmospheric gas chemistry in EAMv3.

Figure 20a shows the difference in stratospheric aerosol optical depth (SAOD) between EAMv2 (based on prescribed sulfate extinction from CMIP6 input data) and EAMv3, which uses sulfur emissions data based on volcanEESM (Ke et al., 2025). The two SAOD data sets are generally aligned well in terms of major volcanic eruption time and intensity, but EAMv3 has a smaller SAOD than EAMv2 during quiescent periods and two additional eruptions between 1940 and 1960. Ke et al. (2025) showed a discernible difference in the impact of the two eruptions on the model simulation of radiation and tropospheric clouds.

Figure 16. Comparisons of zonal mean annual cycle climatology (years 2005–2014) of stratospheric column ozone (SCO, in Dobson units (DU); panels (a)–(d)) and ozone hole (panels (e)–(h)) among observations, EAMv2, and EAMv3. (a) EAMv3 Atmospheric Model Intercomparison Project (AMIP) ensemble mean; (b) Ozone Monitoring Instrument + Microwave Limb Sounder observations (OBS); (c) EAMv3-OBS relative differences (unit: %) of AMIP ensemble mean; (d) annual zonal means; (e-f) time series of the southern hemisphere (SH) minimum total column ozone (unit: DU) and the SH maximum ozone hole area (with total ozone <220 DU, unit: million km^2) from the daily averages of 1 July to 31 December; (g, h) daily mean climatology (lines) and variance (shading) of the SH minimum total column ozone and the SH maximum ozone hole area from years 1990–2014. The red dotted lines indicate 220 DU of total ozone. The shadings represent ±1 standard deviation.

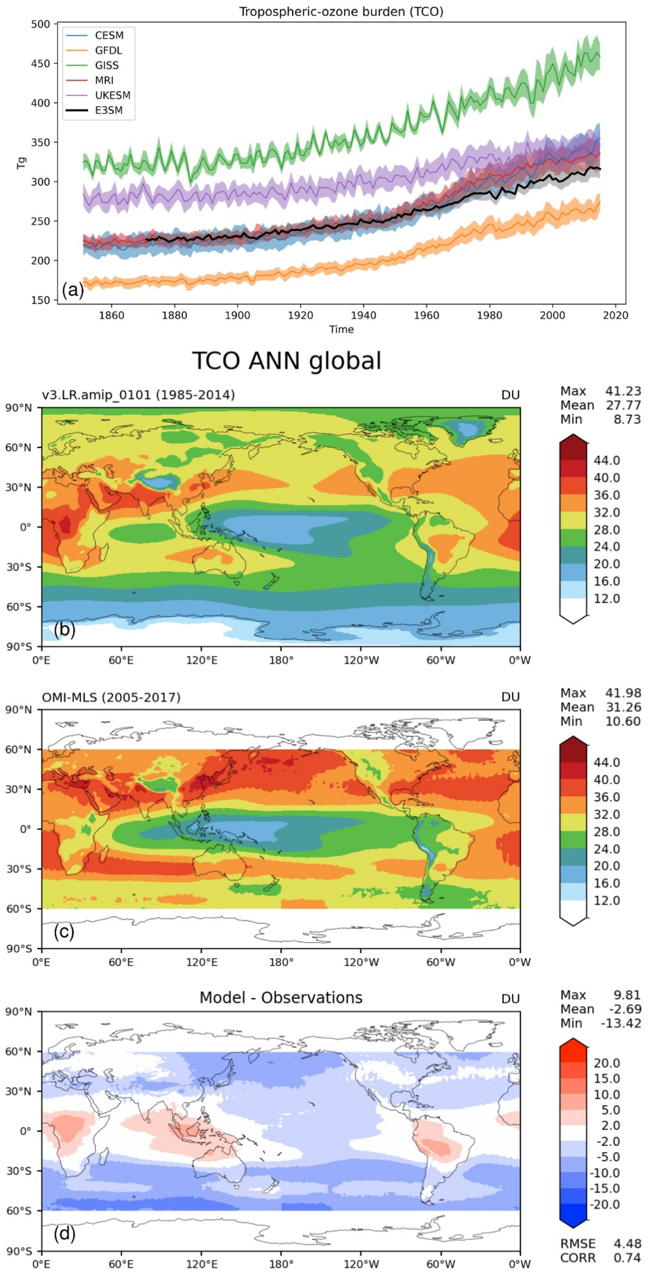

Figure 17. Comparisons of tropospheric ozone among observations, EAMv3, and CMIP6 models. (a) Time series of global annual mean tropospheric ozone burden (unit: Tg). The shading shows the mean ±1 standard deviation of the monthly variability of each year. Tropospheric column ozone (TCO, unit: Dobson units) of annual mean climatology from (b) EAMv3, (c) Ozone Monitoring Instrument + Microwave Limb Sounder observation, and (d) model-observation difference.

The evolution of SAOD after Mt. Pinatubo volcano eruption, evaluated against AVHRR observations from 1991 to 1994 (Figure 20b), suggests that the interactive representation of stratospheric sulfate gives a better simulation of the temporal variation of volcanic sulfate in EAMv3. Hu et al. (2025) also showed that the northward transport of sulfate and the spatial distribution of SAOD are also well captured by the new MAM5-PSA scheme, which has large impact on the radiative forcing distribution of volcanic sulfate. This has been partly attributed to the comprehensive representation of chemical oxidations for the conversion of SO_2 gas to sulfuric acid gas, the nucleation and condensation of sulfuric acid gas to form sulfate aerosol, and the subsequent microphysical processes related to the growth of stratospheric sulfate in MAM5-PSA (Hu et al., 2025).

4.3. Evolution of Aerosol Effective Radiative Forcing in EAMv3 Compared to Previous Model Versions

The total aerosol forcing (ERF_{aer}) and decomposition into its components in EAMv3, following the partitioning method of Ghan (2013), is shown in Figure 21. EAMv3 exhibits a notably weaker negative aerosol effective radiative forcing (ERF_{aer}) than earlier versions of the model. The ERF_{aer} has substantially decreased in magnitude (i.e., become less negative) over the course of EAM's development, from −1.64 W m^{-2} in EAMv1 (C. Zhang et al., 2022; K. Zhang et al., 2022), to −1.33 W m^{-2} in EAMv2, and further down to −0.71 W m^{-2} in EAMv3. This decline in aerosol forcing results primarily from improvements in the model's representation of aerosols, clouds, turbulence, and aerosol-cloud interactions, including both parametric changes and model developments. The aerosol-radiation interaction (ERF_{ari}), and aerosol-cloud interaction (ERF_{aci}) forcing components are −0.77 W m^{-2} and -0.03 W m^{-2}, respectively.

Table 4 summarizes the evolution of key aerosol forcing components, including ERF_{aer}, the aerosol-radiation interaction (ERF_{ari}), and aerosol-cloud interaction (ERF_{aci}), along with comparisons to multi-model benchmarks. The values of ERF_{aer}, as well as its components, ERF_{ari} and ERF_{aci} (−0.03 and −0.77 W m^{-2}, respectively) are largely within the accepted range of the most recent observationally-constrained assessments (Bellouin et al., 2020; Forster et al., 2021; Smith et al., 2021); ERF_{ari} lies slightly outside the central 90%ile confidence interval (−0.71 to −0.14 W m^{-2}) as assessed by Bellouin et al. (2020).

Figure 22 highlights the decomposition of the shortwave component of ERF_{aci} for warm clouds into the ACI radiative forcing (Twomey effect; RF Nd), cloud fraction rapid adjustment (RA, fc), and liquid water path RA (L). E3SMv3 is compared against Aerocom and CMIP5 models, as reported in Gryspeerdt et al. (2020). As evidenced by Figure 22, aerosol-cloud interactions in E3SM v3 are within the range simulated by other models, but they tend to be on the weaker end of that range (i.e., less negative), with the ACI radiative forcing being weaker than all but one of the comparison models. This suppression or reduction of ERF_{aer} ultimately contributed to improving E3SM's fidelity in reproducing the historical temperature record as shown in its coupled historical simulations documented in Part II (Golaz et al., 2025).

Several key EAMv3 developments directly impacted aerosols and their interactions with clouds, including the addition of a convective microphysics parameterization and other updates to the convection scheme, the inclusion of a new VBS-based SOA scheme and several other new or improved aerosol-chemistry capabilities, updates to

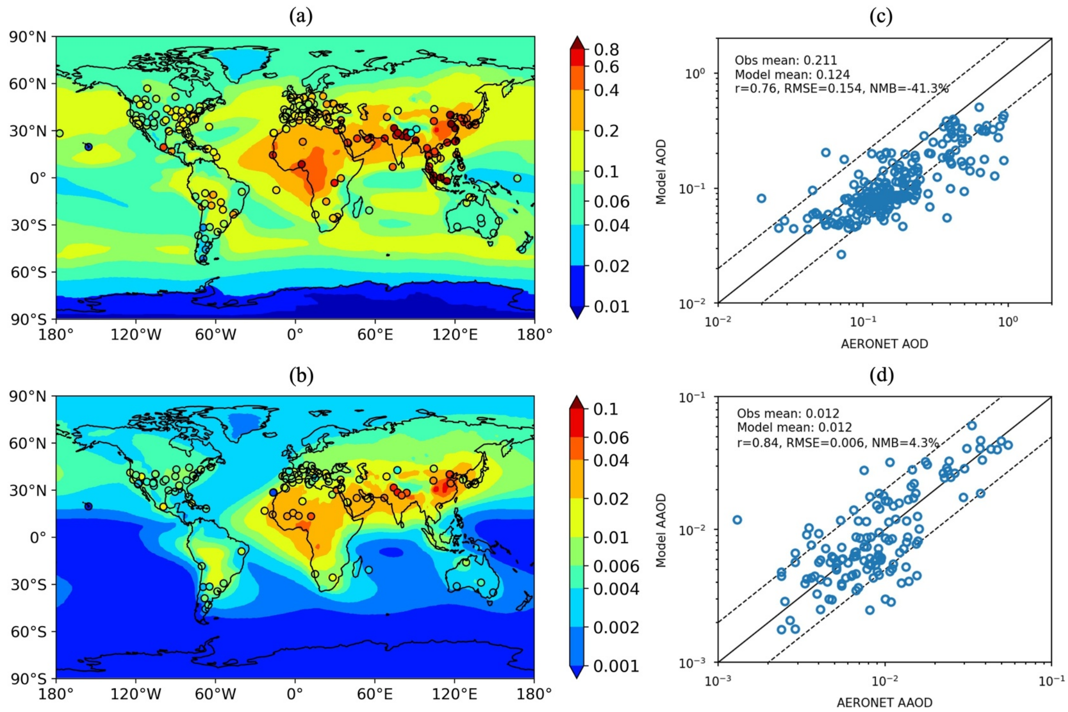

Figure 18. Spatial distribution of the annual mean (a) aerosol optical depth (AOD) and (b) absorption AOD (absorption aerosol optical depth (AAOD)) at 550 nm averaged from 2000 to 2014 based on the EAMv3 (LR.v3.amip) ensemble mean. AERONET surface measurements of AOD and AAOD averaged over 2006–2015 are superimposed in (a) and (b), respectively, as circles filled with corresponding colors. The modeled AOD and AAOD are compared with the observations at the AERONET sites in the scatterplots of panels (c) and (d), respectively, with the solid line and two dashed lines marking 1:1, 1:2, and 2:1 ratios.

aerosol wet removal, and the introduction of a new microphysics scheme P3E (Shan et al., 2024). In addition, several parametric tunings to the cloud microphysics and aerosol schemes also had significant impacts on ERF_{aer}.

Sensitivity analyses allowed for an approximate attribution of individual changes to the net weakening of +0.62 W m^{-2} in ERF_{aer} from EAMv2 to EAMv3, as detailed in Table 5. The three greatest contributors to reductions in (the magnitude of) ERF_{aer} were parametric changes that primarily impacted ERF_{aci}:

1. Increase in minimum Cloud Droplet Number Concentration (CDNC) (Nd) limiter: The prescribed lower bound for Nd was raised from 10 cm to 3 in EAMv2 to 20 cm^{-3}, weakening ERF_{aer} by +0.25 W m^{-2}.
2. Doubling of DMS emissions: This increased background cloud condensation nuclei (CCN) concentrations, weakening ERF_{aer} by +0.12 W m^{-2}.
3. Recalibration of Nd exponent: The autoconversion CDNC exponent was changed from −1.4 to −1.1, weakening ERF_{aer} by +0.1 W m^{-2}.

The increased minCDNC limiter and DMS emissions together comprise about 59% of the net change in ERF_{aer} between EAMv2 and EAMv3. A follow-up paper will further explore the model's sensitivity to these parameters and their impacts on cloud and aerosol properties. It is important to note that the minCDNC limiter is an artificial constraint. Observations in clean environments, such as remote ocean regions, have reported occurrences of CDNC below 10 cm^{-3} (McCoy et al., 2021; Wood et al., 2018). Applying a CDNC limiter of 20 cm^{-3} in the model does not eliminate occurrences of CDNC below this threshold; rather, it reduces the frequency of these low CDNC cases because in the model the mean CDNC values averaged over all sub-timesteps of microphysics calculations, including the timesteps when clouds dissipate, are passed to other model components such as radiation. Clouds with ultra-low CDNC typically have very short lifetimes, because they dissipate efficiently through precipitation. Therefore, adding a minCDNC mainly increases cloud lifetime in the global models, and the effect is larger in the pristine conditions as there are more ultra-low CDNC cases. As a result, the difference in liquid cloud frequency between present-day and pristine conditions is reduced, leading to a weaker aerosol

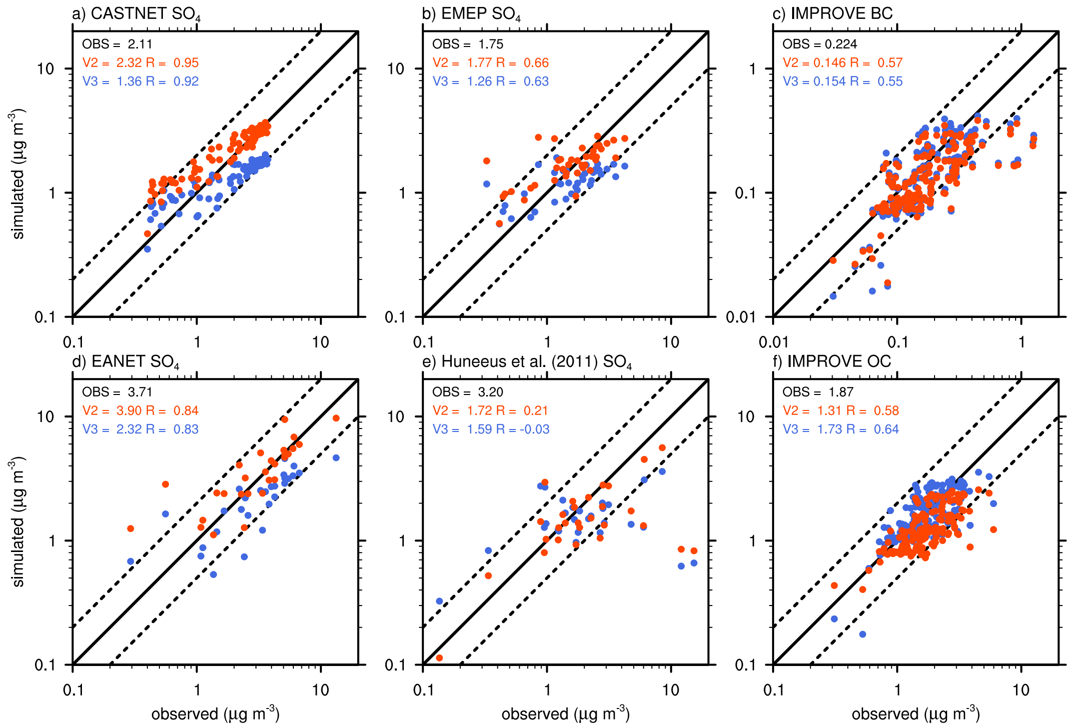

Figure 19. Scatter plots of modeled (EAMv2 in red and EAMv3 in blue) annual mean surface concentrations (μg m^{-3}) of (a, b, d, and e) sulfate, (c) black carbon, and (f) OC aerosols compared to observations from CASTNET, European Monitoring and Evaluation Programme (EMEP), EANET, Huneeus et al. (2011), and Interagency Monitoring of Protected Visual Environments (IMPROVE) surface monitoring sites. Observations at CASTNET, EMEP, EANET, and IMPROVE sites were made during 2005–2014, while observations from Huneeus et al. (2011) were during the 1980s and 1990s. The numbers are mean concentrations and correlation coefficients. The solid line and two dashed lines mark 1:1, 1:2, and 2:1 ratios, respectively.

indirect forcing. Using a 20 cm^{-3} limiter brings the frequency of ultra-low CDNC cases and ERF_{aer} to a more reasonable range.

To calculate DMS emissions, E3SM uses a climatology of monthly mean oceanic DMS concentrations obtained from simulations with a dynamic ocean biogeochemistry model, described in S. Wang et al. (2015). The climatology is consistent across all model years, and the doubling of DMS is applied consistently throughout all PI, PD, and historical simulations. DMS emissions are first calculated by combining the surface concentration climatology with an air-sea flux piston velocity that is a function of wind speed, leading to a global total emission flux of 19.43 Tg S yr^{-1}. In their dynamic model simulation, S. Wang et al. (2015) computed a global oceanic DMS flux to the atmosphere of 20.4 Tg S yr^{-1}; Lana et al. (2011) estimated this flux to have an uncertainty range from 17.6 to 34.4 Tg S yr^{-1}, indicating that our choice to double DMS (38.86 Tg S yr^{-1}) may lead to a slight overestimate of DMS-derived sulfate aerosol. However, doubling DMS emissions leads to significantly improved agreement with observed present-day sulfate aerosol concentrations over the Southern Ocean, where sulfate aerosol is predominantly derived from oceanic DMS.

In addition, it is worth noting that EAMv3, like most current global models, likely underestimates other background pre-industrial CCN sources that have recently been demonstrated to be larger than previously thought, in particular, seeping volcanic aerosol (Jongebloed et al., 2023) and new-particle formation (Zhao et al., 2024). These missing sources are likely to balance any potential slight overestimation in background DMS-derived sulfate aerosol. Clearly, improving both the fundamental understanding and model representation of key background CCN sources remains an important area for future research.

Three additional parametric changes were implemented primarily to address the positive bias in absorbing aerosol concentration and ERF_{ari}. Each of these changes also had an offsetting impact of weakening the negative ERF_{aci}, resulting in a net positive impact on ERF_{aer}:

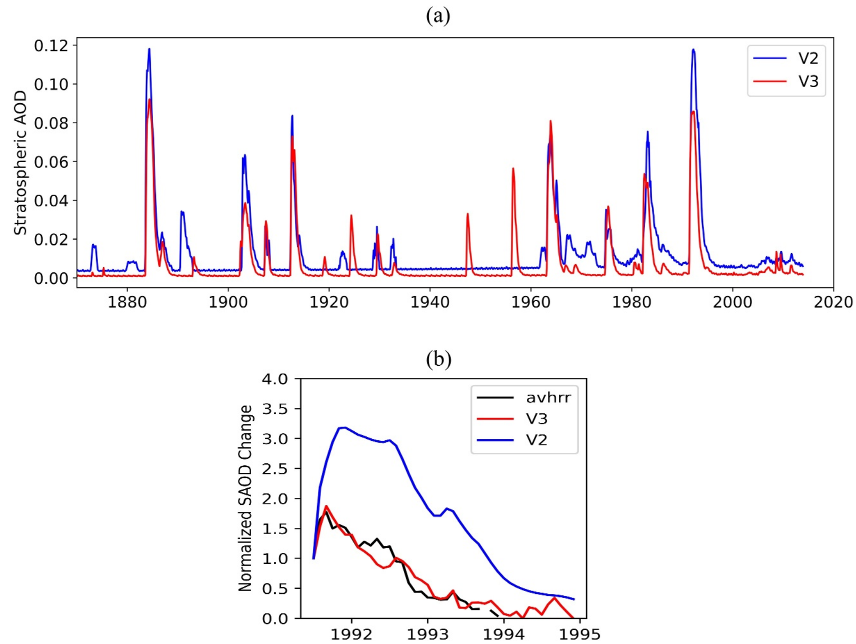

Figure 20. (a) Comparison of simulated historical stratospheric Aerosol Optical Depth (stratospheric aerosol optical depth (SAOD)) between EAMv2 (blue) and EAMv3 (red), and (b) evaluation of EAMv2 (blue) and EAMv3 (red) normalized SAOD against Advanced Very High-Resolution Radiometer satellite retrievals for the Mt. Pinatubo volcano, where normalized SAOD is calculated using their respective values before the eruption (May 1991). SAOD in EAMv2 and EAMv3 is derived by integrating the aerosol extinction from the model simulated tropopause to the model top.

1. Reduction in BC aging lifetime: The number of sulfate or SOA monolayers required for BC aging (Liu et al., 2016) was decreased from 8 to 3, weakening ERF_{aer} by +0.08 W m^{-2}. The updated parameter value is in better agreement with detailed process modeling (Fierce et al., 2025).
2. Increased hygroscopicity of primary organic matter (POM): The hygroscopicity value was raised from near zero to 0.04, weakening ERF_{aer} by +0.08 W m^{-2}. This change reduced BC concentrations, bringing them into better agreement with observed concentrations over the Pacific Ocean. Although the higher hygroscopicity value does not reflect freshly emitted fossil fuel organic matter, it better represents the bulk hygroscopicity of various other POM sources, such as biomass burning aerosol.
3. Wet removal modifications: as described in detail in Shan et al. (2024), several changes were made to the treatment of wet removal, which reduced high biases in anthropogenic aerosol burdens, and thereby increased ERF_{aer} by +0.04 W m^{-2}.

Among the new developments, the largest impact was attributed to the combined effect of the convection updates, including the addition of convective microphysics, the MCSP scheme, and convective mass flux adjustments, strengthening ERF_{aer} by −0.20 W m^{-2} in EAMv3.

Among the new aerosol features in EAMv3, the new SOA treatment was shown to reduce both ERF_{ari} and ERF_{aci}, with a net effect of +0.09 W m^{-2} on ERF_{aer}; the combined impact of the new dust scheme and process coupling order changes was +0.03 W m^{-2}. Additionally, several processes in the model have the potential to impact the effective present-day to pre-industrial aerosol forcing in the E3SMv3 coupled model in ways that are not captured by the ERF_{aer} metric calculations described in Section 3.2.2. These include climate feedbacks onto the emissions of dust and sea spray aerosol via surface wind speeds and soil moisture; the newly-introduced prognostic volcanic sulfate aerosol (e.g., Hu et al., 2025); and the new gas-phase chemistry features, which alter both aerosol concentrations and the fluxes of downwelling and upwelling radiation that interact with aerosols and clouds.

We note that replacing MG2 by P3E weakens ERF_{aer} by only +0.01 W m^{-2} in the final EAMv3.0.0, which is much smaller than the +0.15 W m^{-2} impact shown in Shan et al. (2024). Shan et al. (2024) used an intermediate EAMv3 development version that did not include the new aerosol-chemistry features, nor the other parametric changes listed in Table 4. The impact of P3E on ERF_{aci} is similar in the two model versions (+0.06 W m^{-2} in

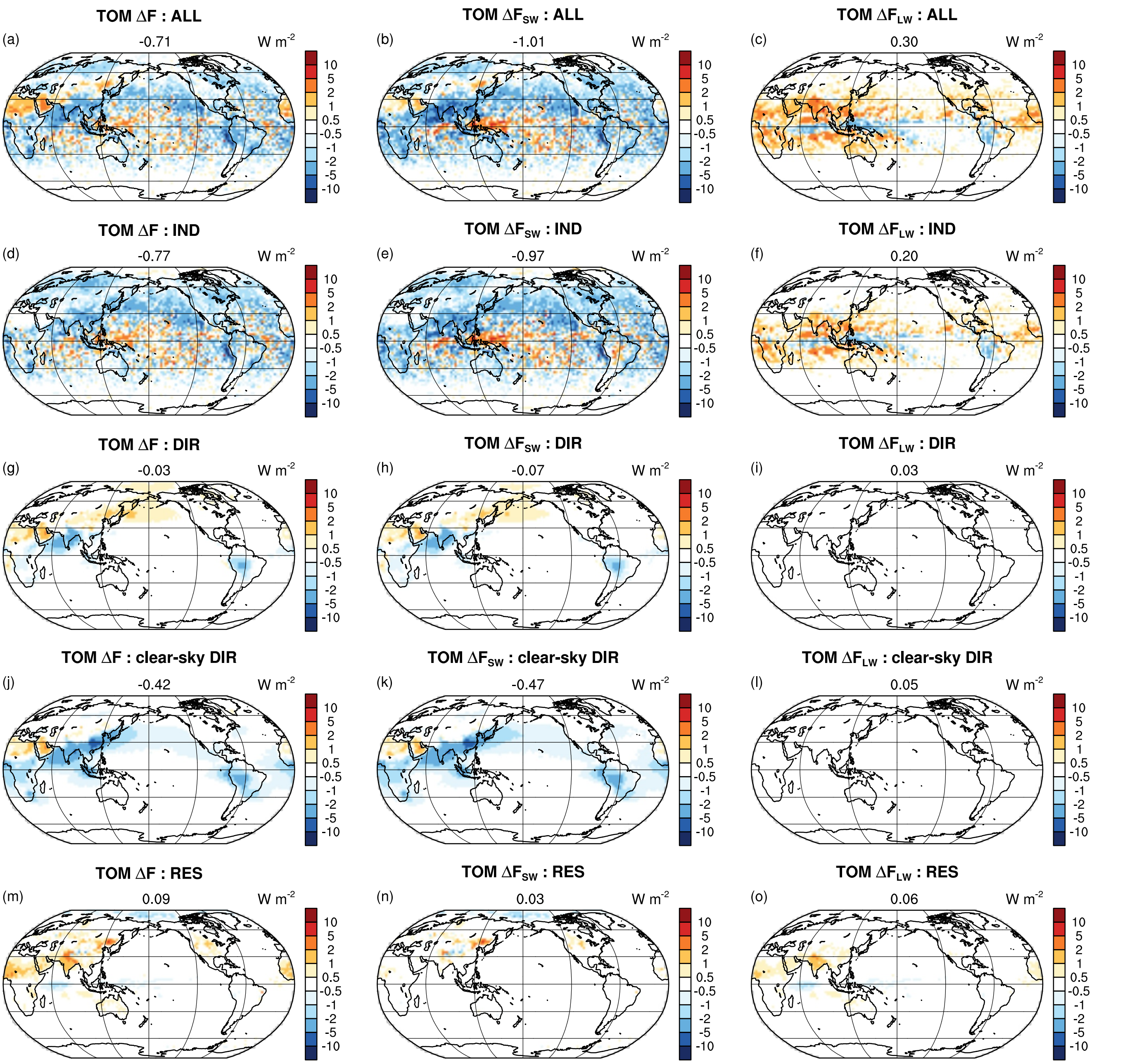

Figure 21. Decomposition of ERF_{aer} in EAMv3 into its components, following the partitioning method of Ghan (2013). The total ERF_{aer} is shown in the top row (panels a–c), and is decomposed into the following terms: aerosol indirect effects (IND; panels d–f), total direct effect (DIR; panels g–i), clear-sky direct effect (clear-sky DIR; panels j–l), and residual term (RES; panels m–o). Each overall term (left column) is further separated into its shortwave (middle column) and longwave components (right column).

EAMv.3.0.0 vs. +0.07 W m^{-2} in Shan et al., 2024), but the impact of P3 on ERF_{ari} is significantly modified following the aerosol-chemistry developments and parametric changes described in Table 4 (−0.05 W m^{-2} in EAMv.3.0.0 vs. +0.10 W m^{-2}), an example of nonlinear interactions among the impacts of multiple model changes on ERF_{aer}. It is also worth noting that P3E substantially reduces both the SW component of ERF_{aer} (from −1.43 with MG2 to −1.02 W m^{-2}) and the LW component (from 0.77 with MG2 to 0.30 W m^{-2}). With MG2 in EAMv3, the magnitude of both the SW and LW components is at the high end among CMIP6 models (Smith et al., 2020), with substantial cancellation of the LW and SW terms. This large SW-LW cancellation is common

Table 4
Evolution of ERF_{aer}, ERF_{ari}, and ERF_{aci} at Key Milestones During E3SM Development, and Comparison Against Benchmark Ranges From Multi-Model Intercomparisons and Community Assessments

	ERF_{aer} (1850 vs. PD, fixed SST), W m^{-2}	ERF_{ari}, W m^{-2}	ERF_{aci}, W m^{-2}	Reference years; summary of key parameters and model changes impacting ERF_{aer}
E3SMv1	−1.64	0.04	−1.77	1850 to 2010; minCDNC = 0
E3SMv2	−1.33	0.04	−1.51	1850 to 2010
				(1) Retuning of turbulence and convection schemes
				(2) minCDNC = 10 cm^{-3}
E3SMv3	−0.71	−0.03	−0.77	1850 to 2010;
				(1) minCDNC = 20 cm^{-3}
				(2) Increased background aerosols by doubling DMS
				(3) Retuned droplet autoconversion
				(4) Faster BC aging
				(5) Increased POM hygroscopicity
				(6) Wet removal improvements
				(7) Additional aerosol features including new VBS SOA treatment
				(8) Convection updates (convective microphysics, MCSP scheme, and mass flux adjustments)
				(9) Change from MG2 to P3E microphysics
Multi-model range (Smith et al., 2020)	−1.01 ± 0.23			1850 to 2014
	−1.37 to −0.63 (range)			
Constraints from multiple lines of evidence (Bellouin et al., 2020)	−1.6 to −0.6 (68%iles)	−0.71 to −0.14 (90%iles)	−2.65 to −0.07 (90%iles)	1850 to 2005–2015 mean
	−2.0 to −0.4 (90%iles)			
Energy balance model with observational constraints (Smith et al., 2021)	−1.1 [−1.8 to −0.5]			1750 to 2005–2014 mean
IPCC AR6 assessment (Forster et al., 2021)	−1.3 [−2.0 to −0.6]	−0.3 [−0.6 to 0.0]	−1.0 [−1.7 to −0.3]	1750 to 2011

Note. E3SMv1 values were originally reported in Zhang et al. (2022).

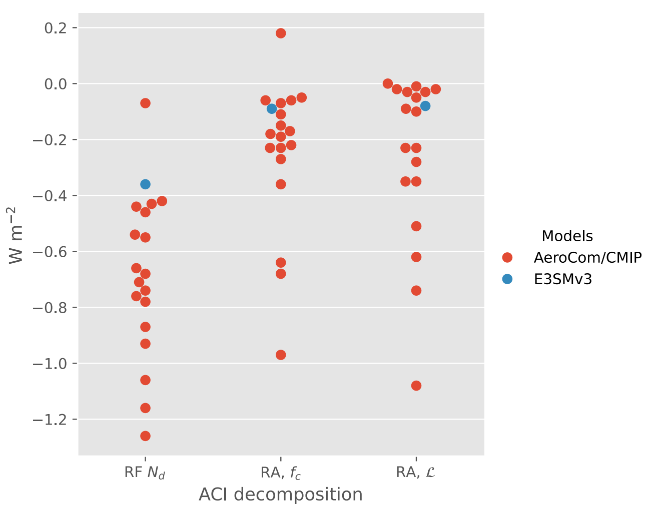

Figure 22. Decomposition of SW ERF_{aci} for warm clouds into the ACI radiative forcing (Twomey effect; RF N_d), cloud fraction rapid adjustment (RA, f_c), and liquid water path RA (L). E3SMv3 is compared against Aerocom and CMIP5 models, as reported in Gryspeerdt et al. (2020).

among models that treat mixed-phase and ice ACI (Gryspeerdt et al., 2020; Heyn et al., 2017; Lohmann, 2017) and is reduced with P3E.

The unattributed residual term of +0.06 W m^{-2} is similar in magnitude to the typical variability between repeated quantifications of ERF_{aer} with differences in machines, compilers, or meteorological nudging data sets (ca. ±0.05 W m^{-2}). Additionally, it is worth noting that nonlinear interactions between individual model changes may contribute to the residuals, as in the example of P3E. Finally, Table 5 is not exhaustive: other model changes, such as the new gas-phase chemistry, could have potentially impacted ERF_{aer}, but these impacts are expected to be minor.

Our findings align with previous studies, such as Hoose et al. (2009), which demonstrated that imposing lower bounds on Nd in GCMs significantly weakens ERFaci, reducing the aerosol indirect effect by up to 80%, while imposing minimum constraints on aerosol had a similar, but weaker impact. Likewise, the adjustments to the Nd exponent in the autoconversion equation mirror the results of Mülmenstädt et al. (2020), who showed similar impacts on aerosol-cloud interactions in the ECHAM model.

Finally, the significant impact of DMS emissions on ERF_{aci} also aligns with prior research. For example, Carslaw et al. (2013) showed that among the

Table 5
Approximate Impacts of Individual Aerosol and Cloud Microphysics Changes on ERF_{aer} Changes From EAMv2 to EAMv3

Model change	Approximate impact on ERF_{aer} from sensitivity tests (W m^{-2})	Percentage of total change in ERF_{aer} (EAMv2 ⇒ EAMv3: +0.62 W m^{-2}) (%)
Increase in minimum Nd limiter from 10 to 20 cm^{-3}	+0.25	40
2 × DMS (with minCDNC = 20 cm^{-3})	+0.12	19
Adjustment of autoconversion N_d exponent from −1.4 to −1.1	+0.10	16
Addition of new VBS SOA scheme	+0.09	15
Reduction in BC aging monolayers parameter from 8 to 3	+0.08	13
Increased hygroscopicity of POM	+0.08	13
Wet removal improvements	+0.04	6
Updates to dust emission scheme and dry deposition process coupling (combined effect)	+0.03	5
Replace MG2 with P3	+0.01	+2
Convection updates bundle: convective cloud microphysics, MCSP scheme, and mass flux adjustments	−0.20	−32
Residual term	+0.06	+10

Note. The impacts of doubling DMS and increasing minCDNC were calculated from one-at-a-time sensitivity tests based on the final EAMv3.0.0 tag, following the methods described in Section 3.2.2. All values were calculated using either the final EAMv3.0.0 or near-final development versions of EAMv3, with results expected to be similar to the final EAMv3.0.0. The + (–) denotes the positive (negative) contribution to the change in ERFaer from EAMv2 to EAMv3.

aerosol-related processes represented in the GLOMAP model, those that controlled natural background aerosol and CCN concentrations, such as volcanic sulfur dioxide, marine DMS, biogenic volatile organic carbon, biomass burning and sea spray, were the primary sources of uncertainty in ERFaer since 1750. And Mulcahy et al. (2018) showed that increasing DMS emissions by a factor of 1.7 led to a change in ERF_{aer} of +0.39 W m^{-2} in the UKESM1 model.

The updates to wet removal, POM hygroscopicity, and BC aging have improved the model's absorbing aerosol burdens as well as reduced the positive bias in ERF_{ari}. These changes simultaneously weakened the (negative) ERF_{aci}, resulting in a total net impact on ERF_{aer} of ca. +0.20 W m^{-2}.

Collectively, these changes resulted in EAMv3 having an ERF_{aer} of −0.71 Wm^{-2}, substantially weaker than in either EAMv1 or EAMv2. This weaker ERF_{aer} contributed significantly to the improved simulation of the historical temperature record in the E3SMv3 coupled model. Simultaneously, these changes contributed to an improved representation of the aerosol life cycle and aerosol geographic and vertical distributions relative to previous model versions, especially for key anthropogenic species impacting ERFaer (BC, POM, and SO4).

The strong sensitivity of ERF_{aer} to the prescribed lower bound on Nd, as well as to natural background sources of CCN, motivates further work to better understand and address the structural causes of extremely low Nd values in the GCM microphysics schemes, as well as the processes that contribute to background CCN concentrations.

5. Computational Performance

The new developments for EAMv3 not only increased the number of processes represented in the model, it also more than doubled the number of tracers from 40 to 82 species and increased the vertical model levels from 72 to 80 layers. This added complexity inevitably increased the computational cost. Tests on 22 and 45 of NERSC's Perlmutter CPU nodes show an increase of 23% and 26% in the model run time of the atmospheric component.

If we look at the individual process contributions to the increased cost in the 45-node simulation (Figure 23), the largest increases in runtime are found in the dynamics and physics/dynamics coupling, which are due to the increase in tracer species that are advected and remapped between the physics and dynamics. In EAMv3, 5 additional tracers are added with Linozv3, 26 with chemUCI, 2 with MAM5, and 8 with the SOA scheme (semi-volatile and intermediate volatility organic gases using the VBS approach). The increased complexity in the chemistry and aerosols, discussed in Sections 2.1.1 and 2.1.2 also increased the cost of the chemistry and aerosol

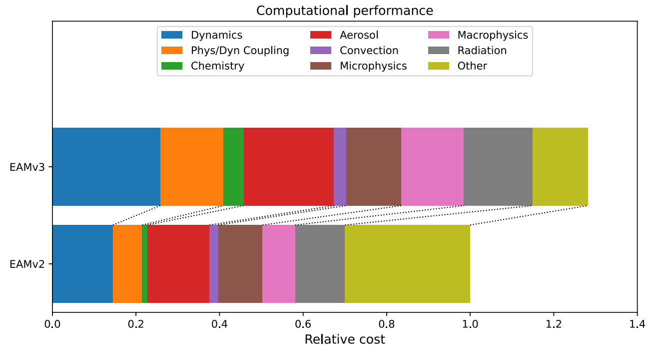

Figure 23. The relative computational cost of EAMv3, compared to EAMv2, and the various atmospheric schemes that contribute the total cost. These computational cost estimates are computed from 2-month 45-node simulations performed on NERSC's Perlmutter CPU nodes. Timings for each of the schemes are obtained from the following timers within the EAM code: Dynamics (a:stepon_run3); Phys/Dyn Coupling (a:stepon_run1+a:stepon_run2); Chemistry (a:chemdr—a:aero_model_gasaerexch); Aerosol (a:microp_aero_run+a:tphysbc_aerosols+a:aero_model_gasaerexch); Convection (a:moist_convection); Microphysics (a:microp_tend); Radiation (a:iation); Other (CPL: ATMRUN—sum of previously listed schemes).

schemes by 10% of EAMv2's atmospheric runtime. The increase in tracers also increased the cost of CLUBB in macrophysics, leading to almost a doubling of its runtime. An increase in the vertical levels from 72 to 80 layers and increasing an additional aerosol mode increased the radiation cost by approximately 40%. The deep-convection enhancements led to a 40% increase in the time spent on convection, but because of its small contribution to the total atmospheric runtime, its relative contribution is small. The slight increase in the microphysics runtime is also small compared to the changes in the other schemes.

To identify the computational cost of increasing the number of model vertical layers from 72 to 80 (Section 2.3.1), we ran two test simulations with a development version of E3SMv3 using 22 and 42 nodes on the NERSC-Perlmutter supercomputer. Based on these results, we estimate that including the eight additional layers increased the overall model cost by approximately 8%–9%.

6. Summary and Future Direction

This paper has provided an overview of the newly released version 3 of the E3SM Atmosphere Model. The model contains substantial updates on the representation of atmospheric physical processes and their interactions. Vertical resolution has been increased from 72 layers to 80 layers to improve the model's ability to capture the QBO. In addition, the number of tracers has more than doubled, from 40 to 82 species, primarily due to the use of the interactive chemUCI gas chemistry. New diagnostics for aerosol and aerosol-cloud interactions, atmospheric chemistry, and clouds were developed and incorporated into the E3SM diagnostics package to support EAMv3 model development and evaluation. The updated atmospheric physical parameterizations are summarized below and their major impacts on simulated climate are summarized in Table 6.

1. The chemUCI interactive chemical module for gas chemistry in the troposphere and an updated linearized ozone scheme for stratospheric chemistry (Linoz v3),
2. A prognostic sulfate aerosol scheme for stratospheric coarse sulfate particles primarily from explosive volcanic eruptions (MAM5-PSA),
3. The MOSAIC module that explicitly simulates the spatiotemporal distribution of nitrate and ammonium aerosols and their radiative effects,
4. An advanced treatment of SOA that includes multigenerational aging through fragmentation and functionalization reactions, particle-phase oligomerization, as well as chemical loss by photolysis,
5. A new dust scheme for calculating the time-varying soil erodibility factor for dust mobilization,
6. Improved aerosol wet removal processes,
7. Enhanced coupling of aerosol-related processes,
8. A new stratiform cloud microphysics scheme P3E to better represent ice physics and aerosol-cloud interactions,
9. A two-moment cloud microphysics for convective clouds,
10. The MCSP scheme for heating associated with mesoscale convection, and
11. Incorporation of the role of the large-scale dynamics in determining cloud base mass flux for the ZM deep convection scheme.

In addition, aerosol and cloud related properties have been significantly adjusted to reduce the overly strong aerosol effective radiative forcing. These adjustments, along with improvements in the wet removal process and cloud microphysics, have considerably reduced aerosol effective radiative forcing, bringing it more in line with community best estimates. As a result, the model's ability to simulate historical temperature records has significantly improved. The improvements in aerosols and chemistry have largely enhanced E3SM's capability to couple aerosol, chemistry, and biogeochemistry for improving projections of future climate changes under various socio-economic scenarios. Changes to cloud and convective processes have led to a more robust MJO and substantial improvement in simulating other aspects of tropical variability, including Kelvin waves, mixed Rossby-gravity waves, eastward inertia-gravity waves, and the QBO. The mean climate simulated by EAMv3 is

Table 6
Summary of New Features Implemented in EAMv3 and Their Impacts on Simulated Climate

New features	Impact on simulated climate
1. chemUCI + Linoz v3	Enables interactive tropospheric chemistry and enhances stratospheric chemistry. Improves simulation of Antarctic ozone hole, stratospheric water vapor, tropospheric ozone and radicals, and greenhouse gas radiative forcing Minor impact on global mean climate states
2. MAM5-PSA	Enables prognostic sulfate aerosol from explosive volcanic eruptions. Improves stratospheric sulfate aerosol and its radiative effects. Minor impact on climate in the troposphere
3. MOSAIC	Adds capability for simulating nitrate aerosol and its radiative forcing (for research only; not part of default configuration due to high computational cost). Minor impact on global mean climate states
4. VBS SOA	Adds an explicit treatment of SOA formation and loss to improve the SOA spatial distribution. Minor impact on global mean climate states
5. New dust scheme	Improves AOD in dust-influenced regions; enhances ice nucleation for polar mixed-phase clouds and iron nutrient supply to oceans. Minor impact on global mean climate states
6. Improved aerosol wet removal	Reduces excessive aerosol burden and AOD, particularly sulfate, as well as overly strong aerosol direct and indirect forcing seen in EAMv2
7. Enhanced coupling of aerosol processes	Addresses earlier issues of excessive dust dry removal, short dust lifetime, and high sensitivity to vertical resolution. Has negligible impact on global mean climate states after re-tuning of dust and sea salt emission factors
8. P3E	Has a large impact on clouds, aerosol-cloud interaction, precipitation, and cloud radiative efforts. Generally enhances cloud radiation forcing over tropical and subtropical oceans and weakens it in the high latitudes in both hemispheres with improvements seen particularly in the Arctic. Improves low-cloud fraction, cold-phase cloud properties, and frequency of heavy precipitation rates and reduces aerosol forcing
9. Convective cloud microphysics	Enables for more sophisticated microphysics in the deep convective scheme. Strengthens tropical variability, improves phase of diurnal cycle of precipitation, and improves mean state precipitation. Increases aerosol forcing (negative impact) and makes the trade wind cumulus region more reflective
10. MCSP	Represents the heating impacts of mesoscale convective systems that are not represented in models with ~100 km grid spacing. Strengthens tropical variability
11. MAdj	Represents the dynamical impacts of the large-scale circulation on deep convection. Strengthens tropical variability and improves diurnal cycle of precipitation
12. Increased vertical resolution	Improves simulation of QBO

largely comparable to EAMv2. Considerable improvements are seen in surface precipitation, AOD, and LWCRE. However, degradations have been noted in SWCRE, precipitable water, and surface wind stress, which are the areas that will need to be further improved in future model development. For instance, as previously mentioned, E3SM is testing a suite of orographic gravity wave drag schemes (Xie et al., 2020) in its E3SMv3 high-resolution configuration. Initial results indicate that these schemes can effectively reduce surface wind bias along mountain ranges.

Despite the progress made in EAMv3, its simulated climate still has notable errors. Long-standing biases in climate models persist, including the dry and warm biases over the central US, weak MJO, and issues with the diurnal cycle of precipitation. Regional biases in precipitation and clouds remain significant compared to observations, such as over the Southern Ocean. These persistent biases are likely related to land and ocean models as well, so might not be corrected by improvement in the atmosphere models alone. In addition, the coupling between various physical processes requires further improvement. The current development efforts for EAMv3 focus on scientifically addressing the low CDNC issue identified in E3SM to remove the need for an artificial minCDNC limiter, updating its orographic drag scheme, as well as improving aerosol-chemistry coupling and ensuring consistency in dust-iron processes across the atmosphere, sea ice, ocean, and land components of E3SM.

To effectively utilize the Graphics Processing Unit (GPU) architectures of DOE exascale computing, future generations of the E3SM atmosphere model, beginning with version 4, will transition from its Fortran-based EAM to a C++-based version (EAMxx). The development strategy for EAMxx will start with the recently

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developed global storm-resolving model configuration, specifically the SCREAM model (Caldwell et al., 2021; Donahue et al., 2024), and incorporate necessary physical parameterizations from EAMv3 and other DOE projects for its low-resolution (~13 km) configurations. This approach aims to create a unified C++-based EAM that can be used for both high- and low-resolution simulations. For the low-resolution EAMxx, key physical processes such as deep convection and gravity waves must be parameterized. The enhanced ZM deep convection scheme, as well as gravity wave parameterizations for orographic waves, convective waves, and frontal gravity waves from EAMv3, will be refactored and rewritten in C++ to integrate into EAMxx. In addition, the chemUCI code will also need to be refactored and rewritten in C++ to ensure efficient GPU performance for EAMxx. Moreover, the Simplified Prescribed Aerosol scheme currently used in SCREAM will be replaced with the MAM4 scheme, which is being refactored as part of the DOE's Enabling Aerosol-Cloud Interactions at Global Convection-Permitting Scales (EAGLES) project (https://eesm.science.energy.gov/projects/enabling-aerosol-cloud-interactions-global-convection-permitting-scales-eagles). Lastly, improving the scale awareness of atmospheric physical parameterizations will be crucial to ensure that EAMxx performs effectively across different spatial and temporal scales.

Conflict of Interest

The authors declare no conflicts of interest relevant to this study.

Data Availability Statement

All model codes may be accessed on the GitHub repository at https://github.com/E3SM-Project/E3SM. A maintenance branch (maint-3.0; https://github.com/E3SM-Project/E3SM/tree/maint-3.0) has been specifically created to reproduce the EAMv3 simulations used in this work. Bit-for-bit results with the original simulations on identical machines will be maintained on that branch for as long as the computing environment supports it. Complete native model output is accessible directly on NERSC at https://portal.nersc.gov/archive/home/projects/e3sm/www/CoupledSystem/E3SMv3/LR for simulation members v3.LR.amip_0101, 0151 and 0201, along with a set of post-processed time series and climatology product at regular 1° × 1° grid ready for producing the diagnostic figures.

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